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Full Length Article

Multi-period, multi-objective optimisation of the Northern Lights and Stella Maris carbon capture and storage chains

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ABSTRACT

A multi-objective optimisation model for CCS chains, aiming to minimise costs and greenhouse gas emissions, considering various transport options is presented. The model builds upon previous work and covers the CCS chain elements after CO_2 is captured, including conditioning, pipeline and batch-wise transportation, intermediate hub storage and injection at CO_2 storage fields. A prospective Life Cycle Inventory is integrated to evaluate emissions from batch-wise transportation. The model is parameterised for accurate estimations based on site-specific characteristics and is implemented in two standalone CCS chains, that are analogues to the Northern Lights project with dominant ship transport and the Stella Maris concept with direct injection ship transport, also incorporating distinct emission profiles, intermediate storage hubs and injection sites. A third implementation combining both chain concepts is implemented. By increasing in a step-wise manner the weight of the emissions in the objective function, the model evaluates cost and emission trade-offs. The optimisation selects costlier wells to minimise emissions and shifts from batch-wise ships to cross-continent pipelines. Chain emissions decrease over time with CO_2 supply increase. Shipping operation dominates emissions, followed by well construction and infrastructure construction. Across all the implementations, the GHG emission intensity of the chain, after CO_2 is captured, ranged from 3.3 to 14.2 %, depending on the concept and transport option adopted and accounting for regional characteristics (i.e., the electricity supply mix per country).

Introduction

Although no binding agreement has been achieved yet, eighty countries declared they would bring improved nationally determined contributions (NDCs) to the COP26, held in Glasgow, UK, and countries responsible for two-fifths of the world's GDP have committed policies to achieve net zero emissions by 2050. The COP26 made some progress on delivering the Paris agreement and evolving the regulatory framework for carbon accounting; however, if all pledges (i.e. NDCs and policies) are fully delivered on time, IEA estimated the global warming could be kept to 1.8 °C, and the lack of long term commitment (i.e. further than 2030) would mean an average temperature increase of 2.4 °C. The COP27, held in Egypt, evolved some elements on carbon accounting, but timid progress was reported on addressing policy gaps (i.e. longer-term NDCs, bolder commitments for emission reduction) and phasing out of fossil fuels for energy transformation processes (KPMG, 2023, 2022, 2020). At COP28 in Dubai, an agreement that signals the beginning of the end of the fossil fuel era laid the ground for a swift, just and equitable transition underpinned by deep emissions cuts and scaled-up finance (UNFCCC, 2023).

Industrial decarbonisation and reshaping energy systems are crucial to mitigate the effects of climate change. A wide range of stakeholders, including think tanks, major oil and gas companies and non-governmental organisations, perform forecasting exercises on which energy mix would be ideal for reducing or eliminating GHG emissions within a specific time horizon (bp, 2023; IEA, 2022; EIA, 2021; Shell, 2021). All forecasts differ on the total amount of energy demand and thus reflect different energy intensities and energy efficiency measures; however, they all consider population growth and increased access to energy. Therefore, it is possible to infer that although fossil fuels are still relevant in the forecasted energy mix, comprehensive implementation of energy efficiency measures, massive deployment of renewable energy and CO_2 mitigation technologies such as CCUS are relevant to sustain GHG emissions at levels required to limit average global temperature increase. Similar pathways for reducing the emissions of GHG are reported in the IPCC 6th assessment report, indicating low carbon

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Fig. 1. Elements of the CCS chain considered in this paper within the green dashed line.

energy as 68 % of the primary energy, in 2050, to limit warming to 1.5 °C, while recognising the relevance of CCS in achieving this goal (IPCC, 2022). The total expected CCS-related emissions abatement by 2050 varied from 4 Gt CO_2 in 2017 to 2.5 Gt CO_2 in 2019, 5.6 Gt CO_2 in 2020, reached 7.6 Gt CO_2 in 2021 and currently stands at 6.2 Gt CO_2 for a net-zero scenario in IEA's world energy outlook reports.

There is a material gap between the current capture rate capacity (i.e. 42.5 Mtpa) and the rate envisaged in 2050 (i.e. 6.2 Gt CO_2 yr^{-1}), according to the 2022 energy forecast from (IEA, 2022) aiming to achieve net-zero emission by 2050. This gap means that CCS based CO_2 abatement rate should grow approximately 146 times in the next 28 years at a compound annual growth rate (CAGR) of 19 %.

The existing CCS chains usually adopt pipelines as a transport option, thus most previously published techno-economic and environmental studies on the CCS chain focus on pipeline network evolution in a multiperiod temporal horizon with either deterministic or stochastic approaches, favouring chains where the pipeline option is the only option available. This approach has been successfully adopted in network evolution studies in the US, UK, and Europe, as can be observed in the CCS projects under-development stage. In cases where there are barriers to the pipeline option (e.g. geography and environmental barriers) and the CO_2 vol to be handled are small (e.g. lower than 0.1 Mtpa), batch-wise transportation (i.e. ships and lorries) could be considered. Although batch-wise transportation brings interesting opportunities to increase the reach of CCS chains while providing flexibility to stationary emitters, it also brings questions regarding the cost, environmental impacts, infrastructure capacity (i.e. ship routes and roads capacities), risks and social acceptance of this transportation option. Within the environmental impacts, GHG emissions during the operation of a batch-wise chain (i.e. energy usage for conditioning and transporting CO_2) could be particularly relevant as they could decrease the absolute amount of CO_2 stored, potentially relying on CO_2 transport in the energy conversion processes that are already amongst the key sources of GHG emissions. Moreover, a model that can consider regional aspects of such a CCS chain while providing optimisation and allowing one to understand potential trade-offs in designing the CCS chains (i.e. demonstrating the implications of the transport option choice on costs and emissions of GHGs) could help achieve the capacity growth rates envisaged in the climate change mitigation targets pledged.

Technologies adopted and chain description

A CCS value chain comprises several elements within an upstream boundary at one or multiple stationary emitters of CO_2 and the downstream boundary at the final sink, which could be one or multiple CO_2 storage sites. This paper considers the CCS chain's conditioning, transport, intermediate storage, and injection and geological storage elements, as illustrated in Fig. 1.

The capture of CO_2 consists of all technologies and chemical processes used to separate CO_2 before or after combustion processes and those chemical processes with an output stream gas composed mainly of CO_2 (e.g., fermentation of biomass). The conditioning element comprises the compression or liquefaction of CO_2, aiming to increase its density for efficient transportation. The CO_2 conditioning element is linked to the transport mode adopted. If transported by pipelines, CO_2 can be compressed to higher pressures than 7.3 MPa (i.e., dense phase fluid). If batch-wise transportation is adopted (i.e., using ships, lorries or trains), it may be cooled down to temperatures near −56.5 °C and pressurised to 0.51 MPa (i.e., close to CO_2 triple point, in liquid phase). The transportation element involves constructing and operating pipelines, ships, and lorries for transporting CO_2. In general, the pipeline mode of transportation is adopted in most operating CCS chains and is commonly used in CO_2 Enhanced Oil Recovery (EOR) projects (Global CCS Institute, 2022). For batch-wise transportation, an intermediate storage facility may be required to allow the continuous flow of CO_2 across the whole chain. Batch-wise transportation often demands one to three times the batch storage capacity to ensure the constant flow of CO_2. Another case would be if a pipeline network of CO_2 is connected to a batch-wise transport mode, then an intermediate storage site will be required along with a conditioning process to allow batch-wise transportation.

The last element of the chain is the injection of CO_2 for geological storage. The injection of CO_2 can be performed directly via a wellhead accessed by a pipeline, via existing oil and gas exploitation platforms, and by stationary vessels or seagoing ships. The geological storage options include saline aquifers, although depleted oil and gas fields are also sinks considered in some projects under early-stage developments (i.e., Porthos project in the Netherlands).

Conditioning of captured CO_2

The conditioning element of the CCS chain aims to increase the density of the CO_2 stream to enable adequate transportation (Aspelund and Jordal, 2007). The conditioning method relates to the transport option adopted downstream of the chain. CO_2 may be compressed above its critical point (>7.39 MPa) for pipeline transportation into a dense phase fluid. However, for batch-wise ship transportation, the CO_2 is expected to be cooled and pressurised to just above the triple point (−56.5 °C, 0.5 MPa) (Paul et al., 2012). Fig. 2 illustrates the physical phase envelope of pure CO_2 and its suitability for transportation options. It also highlights the temperature and pressure work ranges for compression.

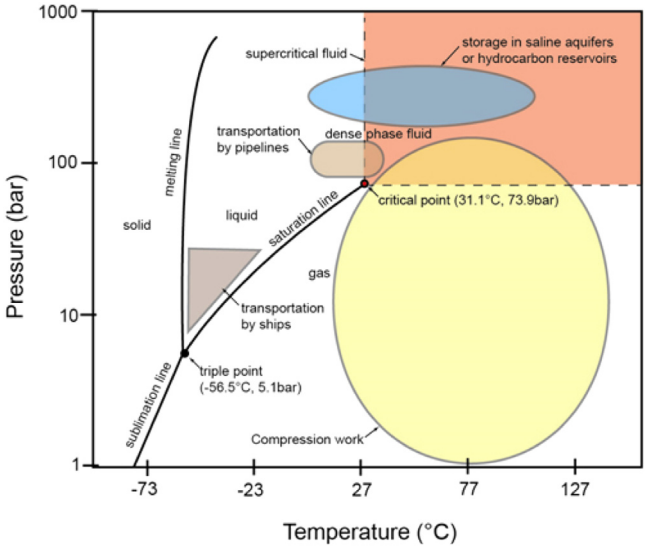


Fig. 2. Phase diagram for CO_2 transport modes and storage (Paul et al., 2012).

Liquefying CO_2 is a typical process at smaller scales than the required for offshore geological storage and is observed in the beverage and food industry and the health sector. The liquefaction process generally comprises compression and intermediate cooling cycles to shift CO_2 back to its isothermal line up to the desired pressure and temperature. The compression stages and cooling cycles are project-specific; however, it is observed in the literature that the number of compression stages varies, generally between three and five stages (Deng et al., 2019; Seo et al., 2015). It is also reported that the outlet pressure selection significantly impacts the total costs, resulting in significant variations in power consumption for liquefaction due to less duty required for cooling on the compression and expansion cycles (Element Energy, 2018). Usually, the liquefaction process configurations fall into two distinct categories, the open and closed cycles. Open cycles contain externally sourced coolant fluids incorporated, whereas closed cycles utilise a refrigerant fluid in a closed loop to exchange heat. Seo et al. (2015) assessed the Life cycle cost (LCC) for four process designs for liquefaction from industry players and theoretical models for 1 Mtpa. The closed system with ammonia as a refrigerant fluid demonstrated the best standalone performance parameter figure of merit (FOM) and the lowest normalised life cycle cost (LCC), despite the higher upfront cost and its complex system nature. The second-best option assessed was the open cycle precooled Lind-Hampson system. A follow-up study integrated the standalone liquefaction systems with the ship-based transport mode for different pressures (Seo et al., 2016). In their conclusion, it is pointed out that, for 0.7 MPa transport pressure, the open system from Linde-Hampson demonstrated the least cost option and that for higher pressures, the closed system results in the least cost option.

Review of batch-wise transport modes for CCS chains

The pipeline transport of CO_2 is the dominant mode, with more than 8000 km length of pipelines installed (Peletiri et al., 2018), and responsible for transporting around 70 Mt CO_2 $year^{-1}$ in projects that are operational (Global CCS Institute, 2021a).

Batch-wise transportation by ships is considered a mature technology at TRL 9 for smaller volumes (up to 1500 m^3) and higher pressures (i.e., 1.5 MPa) than required for large-scale CCS chains (Global CCS Institute, 2021a; Bui et al., 2018). Larger capacities have been proposed but not deployed yet (Neele et al., 2017), and the first ships are being constructed to operate in the commercial scale Northern Lights project; hence its TRL is transitioning to level 8.

Another option for batch-wise transportation is the direct injection from ship aiming to produce relevant flexibility for the chain as it can be deployed at different geological storage sites and does not require an intermediate storage facility for offloading before injection. Currently, its TRL is level 3 (i.e., Proof-of-Concept, component design) and consists of feasibility studies for the Stella Maris concept (Altera Infrastructure, 2020).

Lorries have been suggested for CO_2 transportation from capture pilot plants to other facilities and for shorter distances. One example of using lorries to transport CO_2 occurred in Germany at the Schwarze Pumpe Oxyfuel pilot plant, where CO_2 was transported by lorries to the Ketzin pilot storage site during its operational life. The logistics chain involved 71 trailers, which delivered 1500 t of CO_2 during the test campaign (Burchhardt et al., 2011).

The transportation by ships and lorries may be linked to an intermediate storage hub due to the batch-wise aspect of this transport mode and to enable constant flow and operation of the conditioning process. The storage tanks must be insulated and keep the CO_2 at the conditioned temperatures. Also, investment in pumping, loading and unloading infrastructure is required for ship and lorries transportation. The capacity of the intermediate storage tanks must be at least the size of one load of the transportation mode adopted. For instance, a ship with a 10 kt CO_2 capacity will demand an intermediate storage hub of the same (Element Energy, 2018; House, 2015; Burchhardt et al., 2011).

Ship transportation

Ship transportation of up to 1 kt CO_2 often occurs to supply the food and beverage sectors. Larger scales (e.g., greater than 5 kt CO_2) are suitable for transporting CO_2 captured from power plants and larger stationary emitters to supply a CCS chain. Such a type of ship is not yet operational; however, design options are widely available in the literature. A ship design for a 10 kt CO_2 supporting industry was presented in (Element Energy, 2018), and theoretical design for either optimisation of the CCS chain or standalone analysis of shipping with greater capacity was reported by (Element Energy, 2018; House, 2015; Burchhardt et al., 2011).

The ship capacity designs found in the literature vary from 1 kt CO_2 to 10 kt CO_2, with the latter found most commonly. Higher capacity transport ships (e.g., up to 50 kt CO_2) appear in cost studies, with the primary cost driver being the investment related to building a new ship or retrofitting an existing LPG transportation vessel, with a retrofit being considered to be more economical than a new ship (Element Energy, 2018).

The operating pressure of the vessel is a crucial design parameter for the CCS chain. It impacts processes both upstream (i.e., liquefaction) and downstream (i.e., intermediate storage and offloading) and might result in selecting a new ship or a retrofit from a pressurised vessel. It also can bring incompatibility or higher costs for a transport ship in a broader CCS network if the pressure chosen is incompatible with the infrastructure. The implications of the vessel pressure selection were assessed in (Fraga et al., 2021b; Roussanaly et al., 2021; Element Energy, 2018; CATO, 2016). Those studies mainly explore the trade-offs between higher pressures (i.e., 1.5 MPa) and lower pressure (i.e., 0.7 MPa) transport ships. The former has the advantage of being widely found among the existing fleet (i.e., many years of experience of its adoption for small-scale food-grade CO_2 transportation) and of having lower liquefaction costs (as discussed in §2.1). The disadvantage of higher pressure relies on the lower density of CO_2 at higher pressure (i.e., CO_2 at 1.5 MPa is approximately 13 % less dense than at 0.7 MPa), the additional wall thickness of the storage tanks, resulting in smaller tanks, more inspection space, and higher heel, that ultimately lead to higher material requirements than for the lower-pressure option.

The transfer of liquid CO_2 is done with dedicated loading pumps with loading arms as a more cost-effective alternative to cryogenic pumps with cryogenic hoses. Both could be adopted and are widely used in the LNG industry. Although CO_2 is not considered flammable, its handling has to follow safety procedures such as adding an emergency shutdown (ESD) system. This system is similar to the one used in the LNG industry, with automated ESD valve activation. A second layer of containment to avoid large spills should also be considered. Moreover, the material selection for the transfer lines and elastomers in the seals is crucial due to safety, as loading arms are a high-risk location for leakage (Vermeulen, 2011). Loading pumps with loading arms is the method considered in the FEED report for the Northern Lights project (Equinor, 2020).

The unloading to onshore facilities (i.e., an intermediate storage hub) before further transport to geological storage site(s) or to centralise multiple CO_2 sources is performed by the same infrastructure used for loading, including CO_2 transfer pumps located on the ship. In this operation, the vapour displaced from the storage facility may be returned to the transport ship via a vapour return line on the loading arm. The transport ship should also have pressure relief systems to handle boil-off gas within the ship's storage tanks (Vermeulen, 2011).

The boil-off management within the vessel can be performed through a reliquefaction facility on board or venting to release pressure. The latter has been selected as an option in the Northern Lights project (Equinor, 2020).

Another batch-wise transportation option considered in this research is a direct injection from the ship to the geological storage via a direct connection to the injection wells. This option considers liquid CO_2 is conditioned onboard the ship by heating and compressing it to meet the

reservoir requirements (i.e., adequate temperature and a pressure below the formation fracture pressure). The vessel may be moored to a point or have dynamic positioning systems, allowing the injection of CO_2. The main differences between this option and the transport ship are that the conditioning of CO_2 occurs onboard and that the physical connection from the vessel is to the well (Brownsort, 2015).

The best option tends to be case-specific, as no consensus is found in the literature, and may be driven by the characteristics of the geological storage site and related infrastructure available. For instance, offloading CO_2 to a nearby platform for conditioning and further injection could reduce the equipment on board for both conditioning and mooring, increasing the efficiency of the vessel. This research considers that the direct injection ship has a dynamic positioning system, and conditioning and offloading facilities are installed onboard. This injection ship is considered a generic option and does not rely on the existing infrastructure but can be connected to submerged loading systems and to platforms.

The cost of constructing or retrofitting a vessel is the dominant cost for the capital expenditure (CAPEX) for this batch-wise option. In the case of a direct injection ship, additional equipment is considered (i.e., dynamic positioning, heating, compression, and offloading) and the related costs. In addition, the fuel costs, harbour fees, maintenance and personnel are key drivers for the operational expenditure (OPEX). All those elements are considered for sizing the fleet and trips in a batch-wise dominant CCS chain.

Lorries transportation

Liquefied CO_2 transported by lorries is available commercially in the food and beverage industry and any other sector that demands transport of small quantities of CO_2 (e.g., hospitals). It has also been adopted in pilot projects for the CCS chain (e.g., by Vattenfall at Shawrze Pumpe and at Ketzin Pilot Site in Burchhardt et al. 2011, Martens et al. 2012). This transport option is often neglected in larger-scale projects with greater than 100 ktpa of CO_2 captured for geological storage due to its small capacity (i.e., varying from 9 to 30 t). However, for smaller scales (e.g., lower than 100 ktpa) and shorter distances, even for marginal increases in the capture of CO_2 from emitters close to an intermediate storage hub - which might not justify the expansion of the infrastructure (e.g., new pipeline installation, new ship acquisition) - this transport mode could be considered either temporarily or permanently.

It is worth mentioning that liquefied gas transport by lorries is common practice not only for CO_2 but also for liquefied natural gas (LNG) to supply off-grid areas (Fraga, 2019; Mouette et al., 2019). Also, liquid oxygen is often transported by lorries to hospitals or any industry that demands it in scales that could be supplied by lorries. The main difference between those types of liquified gases and CO_2 is related to density, energy content and the end-use of the liquified gas. Specifically, the currently proposed CCS projects aim for at least 1 Mtpa of CO_2 injection, and further development aims to increase their total injection of CO_2 significantly. As a result, the scale of such projects is incompatible with lorry transportation, given the mismatch between the feasible quantities that can be transported by lorries and the envisaged flow rates of transport and injection for geological storage of CO_2.

However, a lorry sizing and cost model for CO_2 transportation is considered in this paper due to the potential to be used in pilot projects or smaller adjacent networks to a larger chain. Lorry transport also offers flexibility in cases of marginal increase of emission profiles to manage the intermittency of the CO_2 captured and is relevant for CO_2 utilisation at a smaller scale than injection for geological storage and which might not be supported by a permanent infrastructure installation such as pipelines or ship transportation.

Since this technology is not new and is widely used in other industry sectors, cost references can be found in the literature relevant to the freight domain. These are reported as total costs of a service rather than a usual project development, with a business model encompassing one company buying the logistic infrastructure to deliver its CO_2. In other words, CO_2 transport by lorries can be considered a service cost. Nevertheless, new business models can be studied and investment in a logistic infrastructure assessed.

Three types of lorries can be considered based on the existing fleet: heavy distribution containers, articulated containers and tank trucks. Lorries need to be considered regarding the tank size required for the logistics and the condition of the transported CO_2. Similar to ship transportation, tanks must support low CO_2 pressures and be insulated to keep the temperature at liquefied conditions.

Intermediate storage for batch-wise transportation

Storage tanks are necessary elements of a batch-wise value chain to create a buffer between continuous processes, such as liquefaction at the emitter site, and batch-wise processes related to transport by ship. They also enable adjustments on the flow rates for injection, which might need to vary from the flow rates of liquefaction or emission.

The size of the intermediate storage tanks is often considered equal to the capacity of one unit of batch-wise CO_2 transport able to cover at least one round-trip. This relevant parameter may also consider trip delays, weather conditions and unexpected circumstances that could disrupt a trip, impacting the flow assurance of the chain. Estimates from 1 to 1.5 of the batch-wise capacity for intermediate storage capacity are reported in Brownsort (2015). With regard to the design options considered as buffer storage, this research considers pressurised tanks as options for intermediate storage, although direct injection of CO_2 into onshore saline aquifers located immediately below a stationary emitter can be performed. This is because such configurations have raised significant concerns for social acceptance and led to conflict between stakeholders in relation to health and safety regulations at a CCS project proposed at Barenderecht in the Netherlands (Brunsting et al., 2011).

Pressurised tanks for intermediate storage are similar or identical to the ones used for ship transportation, indicating they may need to follow specifications similar to those for liquefied gas ship tankers, resulting in implications regarding sizing the storage tanks for different pressures. For storage tanks expected to operate at higher pressure (i.e., 1.5 MPa), the increased wall thickness, when compared to lower pressure (i.e., 7 MPa), may induce higher costs and higher footprints. Pressure management should be considered to ease the pressure when filling up the tanks and to add gaseous CO_2 to avoid solidification (Element Energy, 2018). Further research could assess how the increased frequency of the loading and unloading of storage tanks inherent to lower pressure configurations could influence the operational expenditure costs and if these might affect the network design and influence operations optimisation.

The footprint of the facility relates mostly to the type and positioning of the storage tanks with the available spherical, cylindrical horizontal or cylindrical vertical options. The typical design observed in CCS chains under development is that of cylindrical tanks vertically positioned. Cylindrical pressure tanks are produced in a simpler manufacturing process than their spherical counterparts, making them potentially more economical to produce and more efficiently placed in rectangular structures (e.g., buildings and ships). Spherical pressurised tanks have the advantage of requiring less wall thickness than cylindrical ones expected to operate at the same internal design pressure and hold a larger amount of CO_2 per unit surface area, resulting in a lower heat transfer surface (Fraga et al., 2021b).

The existence of height limitation in the surrounding space favours the adoption of horizontal tanks. Vertical tanks are favoured by requiring only one connection point to the ground, whereas for horizontal tanks, both fixed and sliding support points are required to accommodate thermal expansion. Vertical tanks have a smaller footprint and thus may be more space-efficient while requiring greater ground stability. Additionally, vertical tanks have the advantage of having an exit to a sump at their lowest point, while horizontal tanks require a

specially-designed sump. Overall, horizontal tanks are the most common and convenient choice, particularly for small volumes such as those used for transport by lorries; however, in settings where space is constrained or for large volumes, vertical tanks are more frequently utilised (Fraga et al., 2021b).

Geological storage

The geological storage of CO_2 is achieved by a combination of structural and stratigraphic, residual CO_2 trapping, solubility trapping, mineral trapping and adsorption trapping mechanisms (Nie, 2009). The first consists of storing CO_2 below low-permeability seals or caprocks such as shales, salt beds and other appropriate formations, which can function as a seal and prevent leakage. The reservoir pressure should not exceed the fracturing pressure to avoid caprock fracturing and fault activation (IPCC, 2005; Streit et al., 2005). When supercritical CO_2 is injected in saline aquifers, its movement acts to displace the formation water and driven by buoyancy moves upwards within the storage formation, reducing the free CO_2 saturation in the zone where it was initially injected, typically at the lowest part of the formation. When the gas saturation drops below the residual gas saturation, the CO_2 does not move and is trapped in the pores, in a process that is referred as residual CO_2 trapping (Thomas, 2005). Solubility trapping occurs with the dissolution of the injected CO_2 in the existing fluids within the pores (e.g., brine, water, hydrocarbons) (Shaw and Bachu, 2002). As a result of chemical reactions in the reservoir, CO_2 can form new minerals regarded as the most stable, permanent form of storage through mineral trapping mechanisms that may take thousands to millions of years to complete (IPCC, 2005).

The most widely practised geological storage method for CO_2 currently is as part of enhanced oil recovery (EOR) and accounts for the majority of CO_2 currently stored in the US. Permanent CO_2 storage in saline aquifers as part of commercial-scale projects has taken place onshore at the In Salah site in Algeria (although not currently operating) and offshore in the North Sea and Barents Sea, in Norway. Enhanced gas recovery (EGR) and permanent storage in depleted oil and gas fields are also considered significant opportunities with an expected increase in TRL level with recent developments announced in the Netherlands, Norway, and Denmark (Danish Energy Authority, 2023; ALIGN, 2019; Bui et al., 2018).

EOR methods are related to the tertiary recovery of oil, which is characterised by the miscibility of the injectant fluid, which can be CO_2, N_2 and hydrocarbon gases, with the oil in place at the reservoir to improve the wettability and increase the oil output of the reservoir. More than 140 EOR projects globally account for 0.35 % of the global daily oil production. Most of the projects are in the US. Part of the injected CO_2, mixed with the oil, is recovered during the exploitation process and separated. This CO_2 can be re-injected to maintain the pressure level at the reservoir or injected into another storage site.

The Sleipner, Snøhvit and In Salah CCS projects have reported feasible injection for permanent storage in saline aquifers since 1996. Reservoir condition varies on the order of magnitude level, with porosities varying from 0.12 to 0.35 and permeabilities greater than 500 md to 1 d The spread of the CO_2 plume within the reservoir has been driven by geological factors and monitored during the injection period using various methods, including seismic techniques, which have helped to improve the understanding of the reservoirs. High-quality monitoring data is essential in increasing the confidence of the storage project and for robust site characterisation, as per previous experiences from the oil and gas industry (Eiken et al., 2011).

The parameters to be considered for costing the injection and storage stage are related to the number of injection wells, storage capacity and concept of injection used. The injection process may be implemented via pipeline, from a floating platform or injection ship or via direct injection from the ship transporting the CO_2 (Ministry of Petroleum and Energy, 2016; Kler et al., 2016).

Prospective attributional LCI for batch-wise transportation of CO_2

As LCA studies on CCS chains focus on CO_2 capture technologies and only a few consider comprehensive bottom-up modelling throughout the whole chain; however, these also primarily focus on pipeline transportation of CO_2. This section contains the prospective gate-to-gate life cycle inventory (LCI) model developed for the batch-wise transportation of CO_2. This work follows the ISO norm as much as possible; however, it cannot be considered a fully ISO compliant LCA study as it focuses on a gate-to-gate boundary. The following sections determine the goal and scope; the inventory modelling and the life cycle assessment carried out, followed by an analysis of the result and discussion.

Goal and scope

Following the ISO 14,040 norm in defining the goal and scope, this analysis aims to provide accurate GHG footprint estimates from the batch-wise transportation of CO_2 using ships and lorries as input for the MILP models developed in this research, so that both CCS chain cost and GHG emissions are considered in the optimisation. The outcomes envisaged in this prospective study are important for stakeholders and developers, as most of the optimisation models for CCS chains do not account for batch-wise transportation and the related GHG emissions are not considered when assessing the whole CCS chain development. Thus, it could provide relevant information for large-scale deployment of CCS chains, particularly in regions where batch-wise transportation plays a significant role (i.e., in Europe). The product system of batch-wise transportation of CO_2 includes liquefying CO_2, intermediate storage, loading, ship or lorry transportation, and offloading of liquefied CO_2. These elements form the product system boundary, comprising the construction and operation phases for a thirty year long time horizon. Instead of a typical functional unit that ought to be focused on the system product, this study uses a reference unit set as 1 t of CO_2 transported as the key reference flow in this gate-to-gate study. Fig. 3 presents the product system boundary, the input of natural resources and energy, the output as emissions and the phases.

The selected impact category for this study is the global warming potential (GWP), and the LCIA methodology used is that of midpoints from CML 2016 with GWP 100 years (Universiteit Leiden, 2016). This study approached the data collection starting from analogue operations (i.e., LNG value chain), then sourced literature data, performed engineering calculations when possible and modelled the findings in Sphera LCA (i.e., former GaBi). This approach is ideal for a prospective study since there is no publicly available evidence of batch-wise transportation of CO_2 operations at a commercial CCS scale, with only pre-feasibility studies based on design and related processes available, with the most mature development from the FEED report from Equinor (2020).

LCI model development

The LCI modelling for the processes indicated in Fig. 3 is individually discussed in this section, and the devised inventory is presented considering the phases indicated in each process (i.e., operation, or construction and operation). The background data and engineering calculations are presented. Overall, the LCI model developed does not consider the decommissioning phase, but in the case of construction, the time horizon described in Fig. 3 is considered. The composition of CO_2 transported that is used in this LCI inventory model is that of the Northern Lights project presented in Table 1. Moreover, the model developed assumes that CO_2 is made available within the specification (i.e., additional purification processes are not required). Further research could expand the system boundary considered in this work to include CO_2 capture, and account for the complexities and possibilities of CO_2 purification processes (e.g., treatment and blending pure and impure streams).

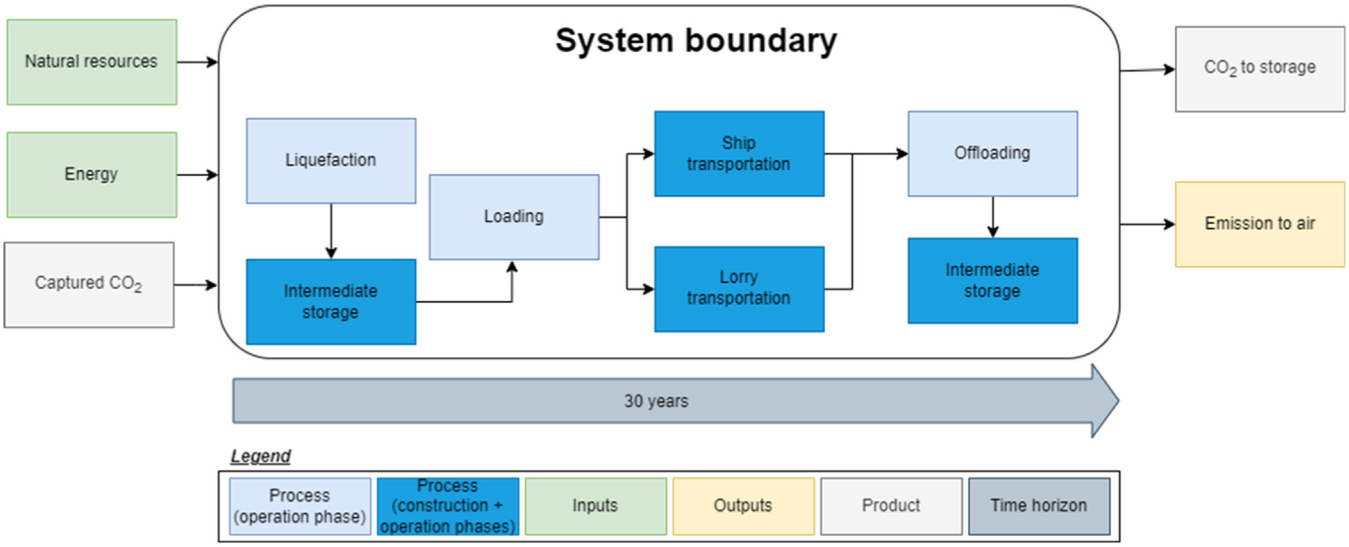


Fig. 3. Product system boundary for the prospective attributional LCA on batch-wise transportation of CO_2.

Table 1
Transported CO_2 composition used in the LCI modelling (Equinor, 2020).

Compound	Value ppm (mol)
H_2O	≤30
NOx	≤10
SOx	≤10
CO	≤100
O_2	≤10
H_2S	≤9
H_2	≤50
Hg	≤0.03

Life cycle inventory model for CO_2 liquefaction

The transportation of CO_2 requires liquefaction to increase its density, which can be achieved through cooling and pressurising just above its triple point. Both main processes for cooling CO_2 (i.e., internal and external cooling loops) have electricity consumption as a crucial parameter for modelling, as are documented in previous studies by Roussanaly et al. (2021), Deng et al. (2019), Seo et al. (2016), which are also considered here.

On the other hand, GHG emissions from the construction of the liquefaction facility, indirect emissions from the production and transport of coolant fluids, and the venting and fugitive (i.e., from valves) GHG emissions have not been considered here as information for these is very scarce and prospective estimates for site-specific emissions which would also depend on the liquefaction method chosen would be too uncertain. In the near future, progress on the Northern Lights project may provide new information regarding the design options and raw materials' requirements, which could be incorporated in future research and for an expanded system boundary.

Specifically, the selection of the liquefaction method considers factors such as the pressure of the captured CO_2, availability of the cooling fluid and feasibility of installing an external refrigeration system. Therefore, this inventory considers two pressure designs (i.e., 7 and 1.5 MPa), the captured CO_2 available at 7 MPa, and the electricity requirement as an average of all the liquefaction reported in the literature for both pressures. Fig. 4 illustrates the boundary of LCI modelled with the elementary flows considered.

The GHG emissions associated with this process are strongly dependent on the location of the facility and the electricity mix associated with this geography (e.g. relative reliance on fossil fuel based and renewable energy). For the specific CCS chains in this research, have used the country specific background inventories for mid-voltage electricity generation. For instance, the Norwegian electricity mix is known to be predominately sourced from hydroelectric generation, thus of a relatively low GHG emission as compared to other countries, e.g. Ireland, where the energy mix relies mostly on coal and NG-fuelled power plants. Table 2 presents the liquefaction inventory model results for each country that is relevant for the MILP models implementation in this research and provides the estimated GHG footprint for the two pressure designs considering the country specific electricity mixes, which are sourced from the GaBi v 2023.1 aggregated dataset.

LCI model for intermediate storage facility construction

Buffer storage using storage tanks is necessary to manage the intermittent nature of batch-wise transportation, enabling the cycle of CO_2 capture and ship loading, unloading while maintaining consistent pressure and temperature during the injection into the geological storage reservoir. The size and capacity of these tanks may vary depending on the facility, batch-wise transport capacity, and desired level of security for the buffer concerning ship roundtrip flexibility. Storage tanks may be situated close to CO_2 stationary emitters and also close to the storage hub, which can consolidate multiple streams of CO_2 captured from stationary emitters, or receive liquid CO_2 from various ship routes, jointly or separately with nearby stationary emitters connected to the hub via pipeline for conditioning and further injection.

The LCI modelling approach devised for this element of the CCS chain considers two distinct contributions to the intermediate storage facility GHG emissions accounting, one from the inventory model for storage tanks and another from the intermediate storage facility construction.

The facility storage tanks can be spherical, vertical, or horizontal cylindrical. The benefit of spherical tanks over the other two types is a thinner wall thickness required and a lower heat transfer surface. The vertical cylindrical reduces the land footprint required compared to the horizontal option, thus is recommended when there is no height limit. For this LCI model, vertical cylindrical tanks are considered for two pressure configurations (i.e., 7 and 1.5 MPa). For the intermediate storage facility, the site preparation, CO_2 import jetty facility, storage of goods and workshop are considered as indicated in Equinor (2020). Fig. 5 illustrates the process boundary and the elementary flows considered for

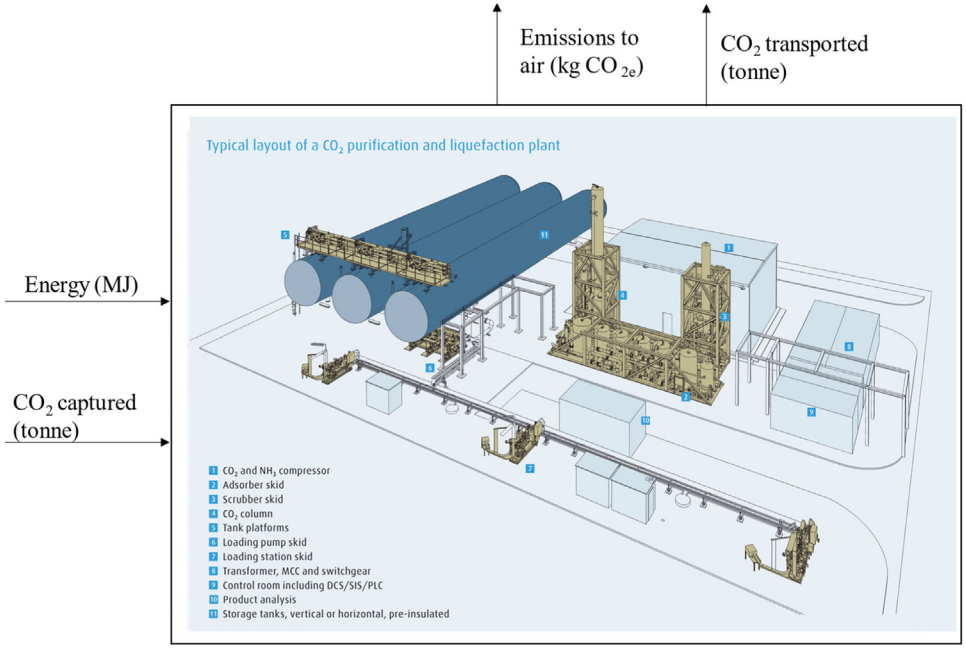


Fig. 4. CO_2 liquefaction processes considered in the LCI model developed after Linde (2017).

Table 2
LCI model results for liquefaction of CO_2.

Pressure design (MPa)	Emissions (kg CO_2e t^{-1} CO_2 transported)				
	Norway	Sweden	Belgium	France	Ireland
0.7	0.9	1.0	4.3	1.8	9.4
1.5	0.7	0.8	3.4	1.4	7.5

Table 3
LCI model parameters for the storage tanks (Fraga et al., 2021b).

Parameter	Unit	Pressure configuration	
		0.7 MPa	1.5 MPa
Maximum thickness of steel	m	0.045	0.045
Density of the CO_2	Kg m^{-3}	1,150	1,000
Allowable stress	MPa	108.6	108.6
Weight of steel A516	t m^{-3}	7,861	7,861
Tank height	m	30	30
Estimated diameter	m	13.96	6.52

the inventory model of the storage tanks, while Fig. 6 shows the same elements for the intermediate storage facility construction.

The design of vertical cylindrical storage tanks has considered the pressure, maximum thickness of steel, the density of the CO_2 under the conditions stored, allowable stress and unitary weight for the steel used as proposed in (Fraga et al., 2021b). Table 3 indicates the parameters adopted for this inventory model for both pressure designs.

Afterwards, the material requirements per storage tank and the number of tanks needed for a particular ship size are calculated and shown in Table 4 for each pressure configuration for one storage tank.

At this point, it is worth highlighting the first implication of different pressure configurations in the material and designing of the intermediate storage tanks. Initially, a low-pressure storage tank (i.e., 0.7 MPa) designed with the wall thickness mentioned in Table 3 can store five times more volume of CO_2 than the 1.5 MPa pressure alternative under the same design parameters. The density of CO_2 at 0.7 MPa is higher than the other pressure configuration, allowing more CO_2 to be stored, requiring fewer materials and energy to operate. Furthermore, the quantity of tanks required for the 1.5 MPa configuration is higher at the lower pressure as the maximum wall thickness of the steel and the stress of the material impose a limitation on the diameter of each tank. Since the sizing of the storage tanks is rounded to the nearest higher integer, it is possible to observe the first two capacities assessed (i.e., 6 kt and 10 kt) share the same number of tanks required for 0.7 MPa pressure configuration. For the 1.5 MPa pressure configuration, the number of tanks is substantially higher when compared with the lower pressure inducing higher amounts of steel required. Overall, the burden for producing the steel for storage tanks is 2.19 tCO_2e t^{-1} steel. Table 5 describes the results for each pressure configuration and ship capacity considered.

Assuming the same materials are used for both pressure configurations, the 1.5 MPa pressure configuration, by requiring more materials, results in higher emissions than the lower pressure configuration.

For the intermediate storage site, the inventory is primarily based on the existing reference from the FEED report from Equinor (2020), which considers the site preparation, infrastructure (i.e., access roads and utilities), CO_2 import jetty for loading and offloading of the ships,

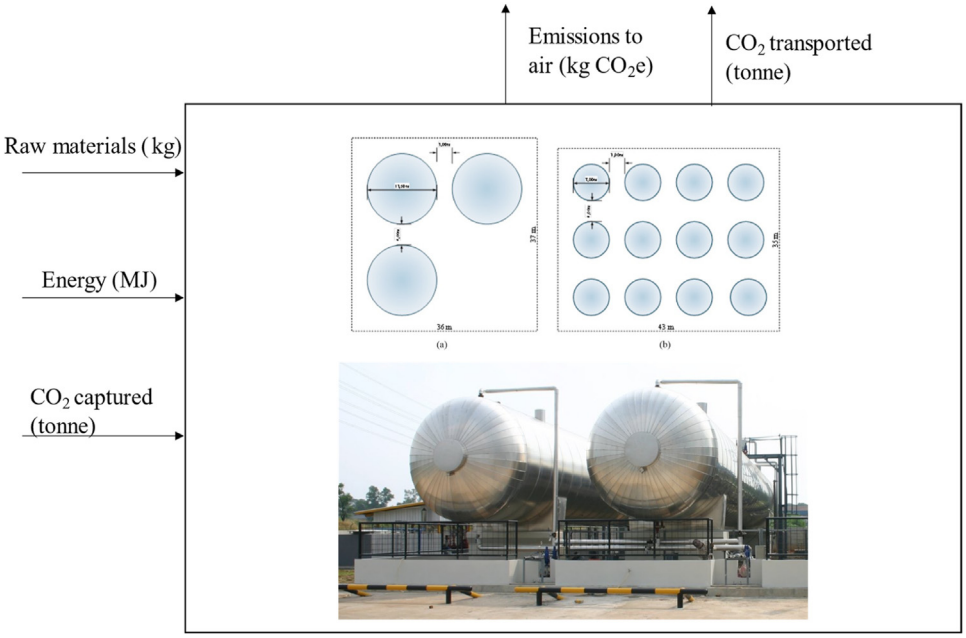


Fig. 5. Process boundary and elementary flows considered for the LCI modelling of storage tanks construction after Pentair (2023) and Fraga et al. (2021b).

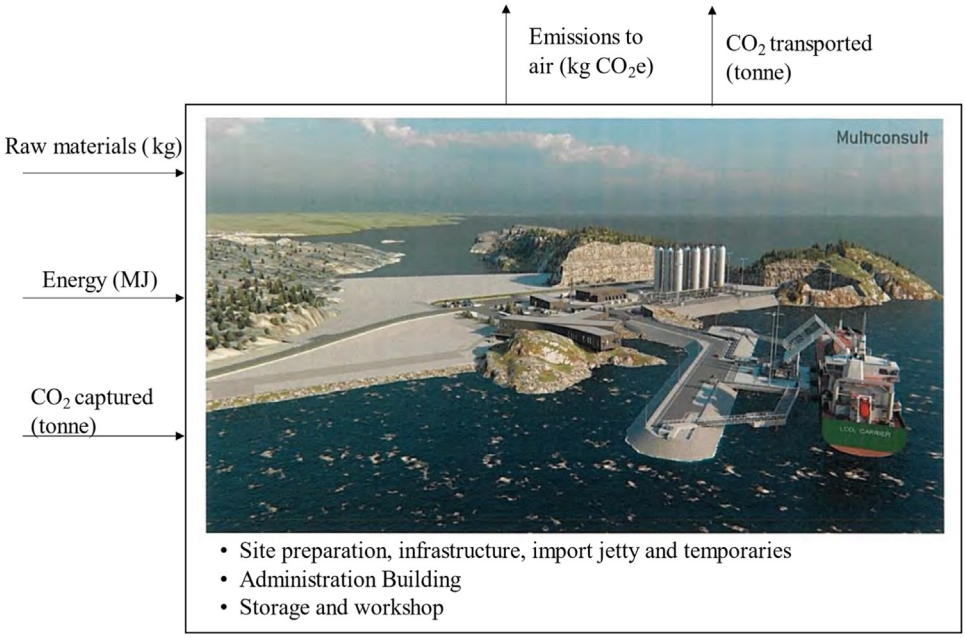


Fig. 6. Process boundary and elementary flows considered for the LCI modelling of the storage facility construction, modified after Equinor (2020).

temporaries (e.g., consumables for the operation of the site), an administration building, and storage and workshop facilities. These elements were designed for an intermediate storage site able to handle 1.5 Mt CO_2, with 12 storage tanks and an import jetty with a 130-metre length. Some of the elements considered should be present in every facility with intermediate storage tanks (i.e., site preparation, jetty for ship loading/offloading, storage and workshop), but some (i.e., administration building) are needed only for intermediate storage hubs. In that sense, table 6 illustrates the inventory adopted for every situation. In the case of elements entirely associated with the intermediate storage tanks, the burden has been linearised based on the ship capacity, thus can be used for all ships using these, and the elements for an intermediate storage hub are considered fixed per site, as would not be directly influenced by the amount of the CO_2 transported to and from the site, but for the operational relevance on concentrating multiple CO_2 flow streams. The linearisation considered a ship with a 7,500 t CO_2 transport capacity designed for the Northern Lights project.

This inventory modelling considers the fugitive emissions of CO_2 as an emission source for the operations of the intermediate storage site. These emissions are from CO_2 escaping valves and during loading and offloading operations and are 0.00286 t CO_2e hr^{-1} per site, as indicated in Nie (2009).

Batch-wise transportation construction and operation LCI modelling

The LCI model for batch-wise transport encompasses ship and lorries transport modes and the associated loading and offloading aspects. Both options consider the construction and operation phases of their life cycle. The maintenance and decommissioning phases are not considered due to lack of data and the significant uncertainties in the corresponding prospecting accounting for these.

Table 4
LCI model results for quantifying steel for storage tanks after Fraga et al. (2021b).

Parameter	Unit	Pressure configuration	
		0.7 MPa	1.5 MPa
Volume of the tank	m^3	4,592	1,000
Steel surface area	m^2	1,621	681
Volume of steel per tank	m^3	73	30,6
Weight of an empty tank	t	574	241
CO_2 stored per tank	kt	5.28	1

Table 5
LCI model results for the construction of intermediate storage facility tanks.

Ship Capacity (kt)	Number of tanks		Steel required (t)		Emissions (t CO_2e facility^{-1})	
	0.7 MPa	1.5 MPa	0.7 MPa	1.5 MPa	0.7 MPa	1.5 MPa
6	2	6	1,148	1,443	2,513	Not applicable
10	2	10	1,148	2,405	2,513	5,268
20	4	20	2,295	4,811	5,027	Not applicable
50	10	50	5,739	12,027	12,567	

Table 6
LCI model results for storage tanks facility construction.

Element	Value	Unit
Site preparation, import jetty, infrastructure, and temporaries	537.06	**t CO_2e kt^{-1} CO_2 capacity**
Storage and workshop	230.53	**t CO_2e kt^{-1} CO_2 capacity**
Administration building	7,067	**t CO_2e facility^{-1}**

Ship construction. There are many designs of CO_2 transport ships, but none are currently operating in the CCS chain. The use of ships for food-grade CO_2 exists and is widely spread but for smaller volumes than the ones required for the CCS chain. Also, there are differences and implications on the pressure design of those ships regarding their transport capacity and capacity to carry other products such as LPG. Furthermore, there is available literature on life cycle studies for existing large-capacity ships that are in operation for other transport sectors.

The inventory model devised in this research considers two existing life cycle Inventories for the ship construction and adds to these the CO_2 storage tank inventory. Consolidating those inventories provides a prospective approximation of the burden of constructing ships for CO_2 transportation at the CCS scale.

The hull construction is divided into two sub-systems called materials and protection. The first system encompasses steel production, processing steel (i.e., welding and cutting), and abrasive blasting for surface finishing or paint removal. The latter approaches the coating and the production of zinc anodes to provide cathodic protection for the hull and the ship's ballast tanks. The second system considered relates to the machinery construction, including the materials and process for constructing one engine and the related auxiliary engines. This inventory was derived from existing LCA databases from Chatzinikolaou and Ventikos (2015). The last element in system relates to the CO_2 storage tanks, which are added to the ship using the inventory presented in § 3.2.2. Fig. 7 illustrates the processes included in this prospective inventory.

An inventory for hull and machinery construction based on emission factors from existing literature, primarily ECO-indicator 99, but also considering manufacturer'' datasets in the case of coating and other elements is proposed in Chatzinikolaou and Ventikos (2015). The LCI model developed was implemented in the Panamax oil tanker with the main parameters described in Table 7, considering one main engine and three auxiliary engines.

Table 7
Panamax oil tanker parameters used in the ship LCI model after Chatzinikolaou & Ventikos (2015).

Parameter	Value	Unit
DWT	74,296	t
Service speed	14	Knots
Length	219	m
Breadth	32.24	m
Depth	20.60	m
Draught	14.17	m
Main engine	368	t
Auxiliary engine	19.7	t

Table 8
Emissions to air from LCI ship construction model applied to Panamax oil tanker.

Emissions to air	Value	Unit
CO_2	2.29 10^4	t
CH_4	4.06	t

In this implementation, the ship construction emissions of CO_2, NOx, all PM, and SO_2 accounted respectively for 2.08 %, 0.42 %,0.92 %, and 0.64 % of the emissions from the whole ship life cycle (i.e., including operation, maintenance, and dismantling). On the other hand, the emissions of CO and CH_4 were, respectively 10.11 % and 78.58 % of the overall emissions. The most significant emission source from the shipbuilding stage was from the hull construction process (e.g., approximately 70 % of CO_2 emissions), primarily driven by steel production (e.g., approximately 96 % of CO_2 emissions) (Chatzinikolaou and Ventikos, 2015). Table 8 below summarises the emissions results for the construction stage of the Panamax oil tanker. Out if these only the GHG emissions have been used for the research reported here.

In addition to the emissions reported in Chatzinikolaou and Ventikos (2015), three emissions from a standard database from GaBi are considered in this inventory model with all the material flows for an ocean bulk carrier/tanker, similarly to the previous study; although, it does not consider ship manufacturing. This inventory is also used as it provides emissions related to the materials required for the ship assembly rather than the overall shipbuilding process. Table 9 summarises these results from the GaBi (v 2023.1) database.

With this result, applying the LCIA method from CML version 2016 and consolidating all the emissions to air in GWP for 100 years is carried out. Moreover, linear regression is applied to determine the GWP

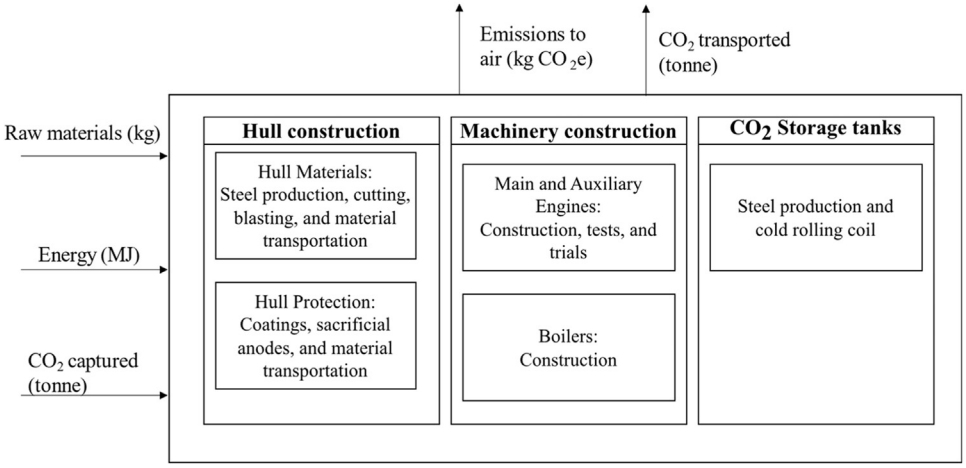


Fig. 7. Process considered in the LCI model of ship construction for CO_2 transport.

Table 9
Emissions to air from LCI ship construction model from GaBi.

Emissions to air	DWT (t)			Unit
	50,000	100,000	200,0000	
CO_2	2.27 10^4	3.71 10^4	6.60 10^4	t
CH_4	1.97 10	3.2 10	5.7 10	t

Table 10
GWP indicator estimates for the LCI ship construction model adopted in this research.

CO_2 transport ship capacity (t)	Estimated DWT (t)	GWP (t CO_2e $ship^{-1}$)				
		Per ship	0.7 MPa		1.5 MPa	
			Storage tank	Total	Storage tank	Total
6,000	7,059	2,876	1,427	4,303	Not applicable	
10,000	11,765	4,793	2,380	7,173	5,268	10,061
20,000	23,529	9,587	4,759	14,346	Not applicable	
50,000	58,824	23,968	11,899	35,867		

per deadweight tonnage (DWT) so the emission from the shipbuilding for CCS could be accounted as a function of the size of the ship. Applying this function to the estimated DWT per ship capacity considered in this research, the resultant GWP indicator can be obtained, and by adding the corresponding value for the storage tanks, considered to be equal tonnage to the tanks devised for the storage facility, the total construction GWP per ship is reached. The estimate of DWT assumes 15 % of this is due to fuel, crew, consumables, and ballast weight. Table 10 summarises the result for each ship capacity considered in this research.

Ship operation. The operation stage for batch-wise transportation by ship consists of fuel combustion for providing propulsion to the ship from its origin to its destination, denoted in the models developed in this research as a trip. A ship trip has moments in which the ship requires the full power from its engine to generate momentum, and when the ship is cruising, demanding less power from the engines. Also, much power is required for manoeuvring and berthing the ship at both ends of the trip. The operation stage is reported to be the most relevant in the LCA of ship transportation. Chatzinikolaou and Ventikos (2015) indicates it can account for the majority of CO_2 (96 %), NOX (99 %), PM (98 %), SO_2 (98 %), and CO (71 %) emissions from the total life cycle of the ship, including construction, operation, maintenance, dismantling. An ISO compliant peer-reviewed LCA for using fossil fuels in the marine sector was developed by Thinkstep (2019). The study aimed to identify GHG emissions from LNG, heavy fuel oil, and marine gas oil. It also considered the fuels since the IMO sulphur regulation, established in 2020, enforcing a lower sulphur content for heavy fuel oil and marine gas oil. The system boundary adopted was from well-to-tank and well-to-wake, as illustrated in Fig. 8 for LNG and Fig. 9 for oil based marine fuel.

The study from Thinkstep (2019) indicated that using LNG as marine fuel could reduce GHG emissions by 6 % to 22 % compared to oil-based fuel post IMO low sulphur regulation. The range depends on the engine considered for the ship, with the 2-stroke engines providing higher reduction than 4-stroke engines.

In terms of shipping fuel for CCS chains, the indication from the FEED report by Equinor (2020) suggests the ships will be operated with LNG with the engine based on state-of-the-art 4-stroke Wartsila 31DF. It also considers battery propulsion and shore power connection for manoeuvring and berthing on the coast could lead to up to a 1.5 % reduction in the CO_2 emissions from the operation stage, improving the local air quality, and is subject to economic feasibility which yet to be proven.

The LCI model considered for this research is for a 4-stroke medium-speed engine, as suggested for the Northern Lights project by Equinor (2020). The emissions associated with the combustion of LNG (i.e., tank to wake) for this engine are adopted from Thinkstep (2019) and summarised in Table 11.

The resultant GWP for this engine is 629 g CO_2e kWh^{-1}, with 488 g CO_2e kWh^{-1} resulting from the combustion of LNG and 141 g CO_2e kWh^{-1} from the LNG supply chain.

Lorry construction. The transportation of CO_2 by lorries is a regular operation as CO_2 is widely used in the food and beverage industry, chemical plants, and hospitals, to name a few uses. Also, liquid gases such as liquid oxygen, LPG, and LNG are widely transported by lorries. These ex-

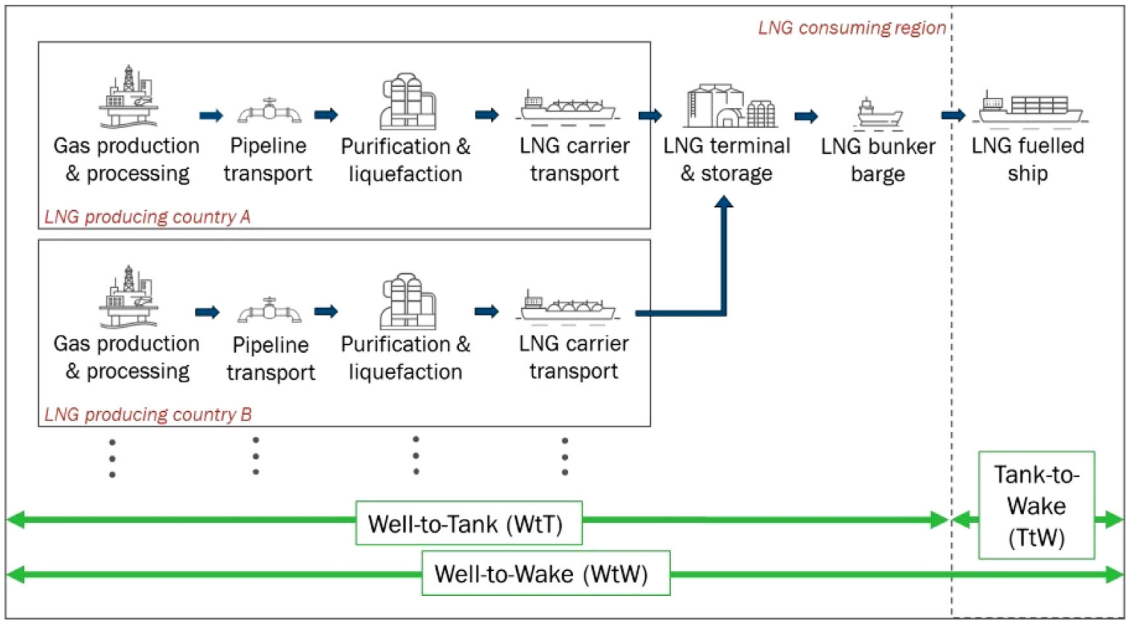


Fig. 8. System boundary for LCA study for LNG maritime fuel after Thinkstep (2019).

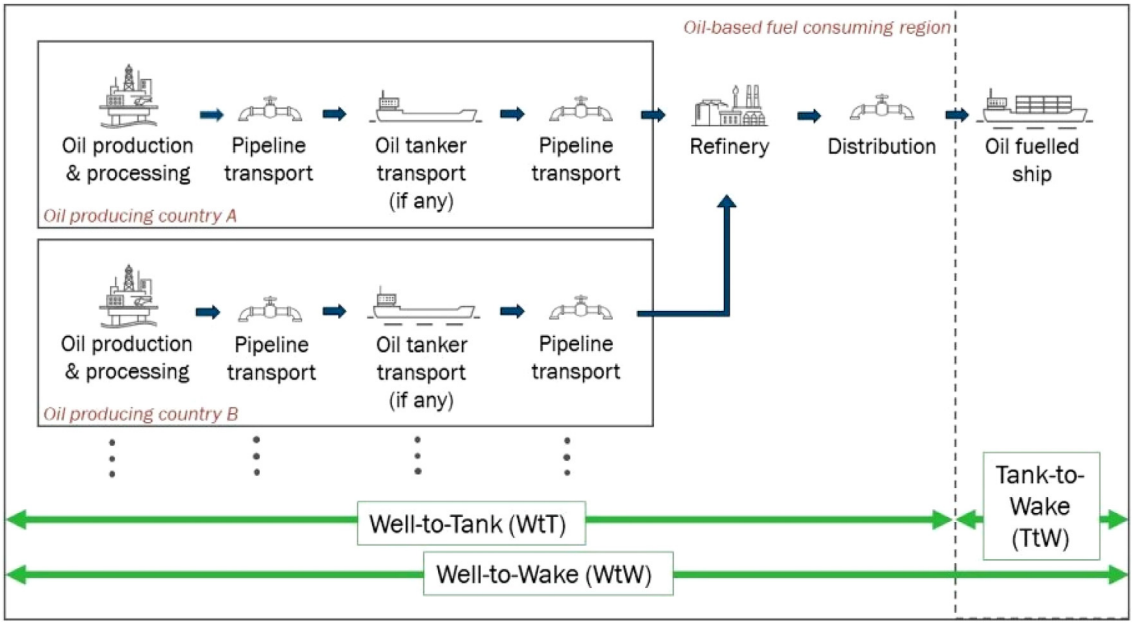


Fig. 9. System boundary for LCA study for oil maritime fuel after Thinkstep (2019).

Table 11
Emissions to air from LNG-fuelled ship operation after Thinkstep (2019).

Source	Value (g kWh^{-1})
LNG consumption	159.3
CH_4	3.84
N_2O	0.017

isting operations provide the technology for the tanks and lorries available for CO_2 transportation; however, as in the case of ships, the cargo size is larger than it usually occurs in current applications.

This research considers that lorries can transport 30 t CO_2, similar to what is designed for the Northern Lights project. The CO_2 from the waste plant in this chain is to be transported by lorries (i.e., fuelled by biomethane produced in the waste plant) to the harbour of Oslo for further transportation to the injection site. In a personal communication with the director of the project for CCS from the waste plant, motivation for adopting lorries as a transportation mode to the harbour in Oslo was the strict regulation which would not allow a pipeline under the hill that is in between the waste plant and the harbour.

Moreover, the length of the investment payback, greater than ten years, was higher than the government support. These conditions are not uniquely found at this particular waste plant and could occur within several CCS chains. Furthermore, this transport mode could increase the extent of the CO_2 transport network and maximise the capture of CO_2 from stationary emitters.

The LCA for the lorries transportation devised in this research accounts for the lorry (i.e., tractor with trailer) and storage tank construc-

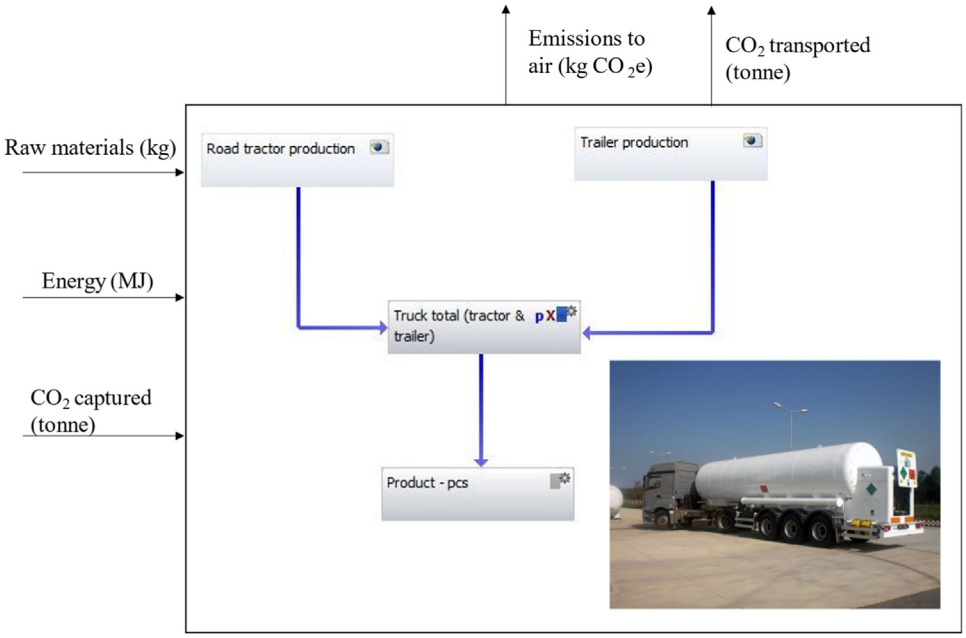


Fig. 10. System boundary for lorry construction for CO_2 transport adapted from GaBi v 2023.1.

Table 12
LCI model parameters for lorry operations adapted from Shires & Loughran (2021).

Parameter	Value (g km^{-1})	GWP (kg CO_2e km^{-1})
CO_2	7.78	7.78
CH_4	3.37 10^{-5}	9.44 10^{-4}
N_2O	2.25 10^{-7}	5.96 10^{-5}
Total	**7.78**	**7.78**

Table 13
LCI model for batch-wise loading and offloading of CO_2.

Transport mode	CO_2 Capacity (t)	Emissions from loading and unloading (kgCO_2e $trip^{-1}$)
Ship	6,000	75.1
	10,000	125.2
	20,000	250.4
	50,000	626.2
Lorry	30	0.3

tion. The former adopts an existing construction process for a suitable lorry from the GaBi database v 2023.1, and the latter considers the same assumptions adopted for the intermediate storage tanks. Fig. 10 illustrates the unit processes considered in the construction of lorries.

The resultant GWP for constructing a lorry to carry 30 t CO_2 is 25.1 t CO_2e $lorry^{-1}$. The burden from the storage tank for the CO_2 must be added. According to the method presented in § 3.2.2, this storage tank accounts for 44.9 t CO_2e $tank^{-1}$. The consolidated GWP figure for the lorry with the storage tank is 70 t CO_2e $lorry^{-1}$.

Lorry operation. In the case of the Northern Lights project, the lorries are fuelled with biomethane, which provides a lower burden when compared to fossil natural gas. Since this might not always be the case, this research considers the combustion of natural gas as fuel for lorries. The GWP is obtained with engineering calculations proposed by Shires and Loughran (2021). The resultant GWP for running a lorry is 7.78 kg CO_2e km^{-1}. Table 12 presents the share of each emission compound considered in the GWP estimate for lorry operation.

Boil-off management. Boil-off management of liquid CO_2 is similar to that of LNG boil-off management techniques. CO_2 is stored and transported in liquid form, but due to heat leaks into the tanks, some CO_2 vaporises and turns into gas, known as boil-off gas (BOG). The BOG must be managed and controlled to ensure that CO_2 storage and transport operations run smoothly and efficiently. There are several ways to manage boil-off in liquid CO_2. Recompression, injection and venting can be considered for BOG management, with recompression being the most common method.

During the intermediate storage tank BOG build-up, the vaporised CO_2 is assumed to be recirculated to the liquefaction process and pumped to the storage tanks, making it a closed loop. In the case of transportation, since reliquefying CO_2 can be energy intensive and the batch-wise transportation mode operates aiming to have high efficiency in its fuels usage, it is assumed that the approach proposed by Equinor (2020) is used, where the tanks are not filled integrally (i.e., up to 98 %) to allow pressure build-up before offloading. In addition, for trips longer than forty-six days, as indicated in Brevik Engineering AS (2020), it is assumed that CO_2 is vented at a rate of 0.2 % of the cargo volume per day (IEAGHG, 2004).

Loading and offloading of CO_2. The loading and offloading of CO_2 from batch-wise transportation are accounted for in this LCI, considering the energy required for the pumps to perform the work. While in the LNG sector, loading and offloading use power from the combustion of fossil fuels, in the CCS chains, they will be powered by electricity from the grid, as indicated in the FEED report from Equinor (2020). It is considered that 0.2 kWh is consumed per t of CO_2 moved during the loading and unloading stages (Aspelund et al., 2006). Table 13 summarises the emission factor for the loading and unloading for different transport options, considering the current Norwegian electricity emission factor of 0.03 kg CO_2e kWh^{-1} from GaBi 2023.1 database.

Results and discussion

The LCI model developed in this research has been used to produce an impact category score for GWP considering a generic application, as described in Eq. (1), comprising the activity and emission factors (i.e., estimated using the LCI model developed).

$$Mass\ CO_2e\ emitted = Emission\ factor \times \left(\frac{t\ CO_2e}{activity}\right) \times \text{Activity } factor \quad (1)$$

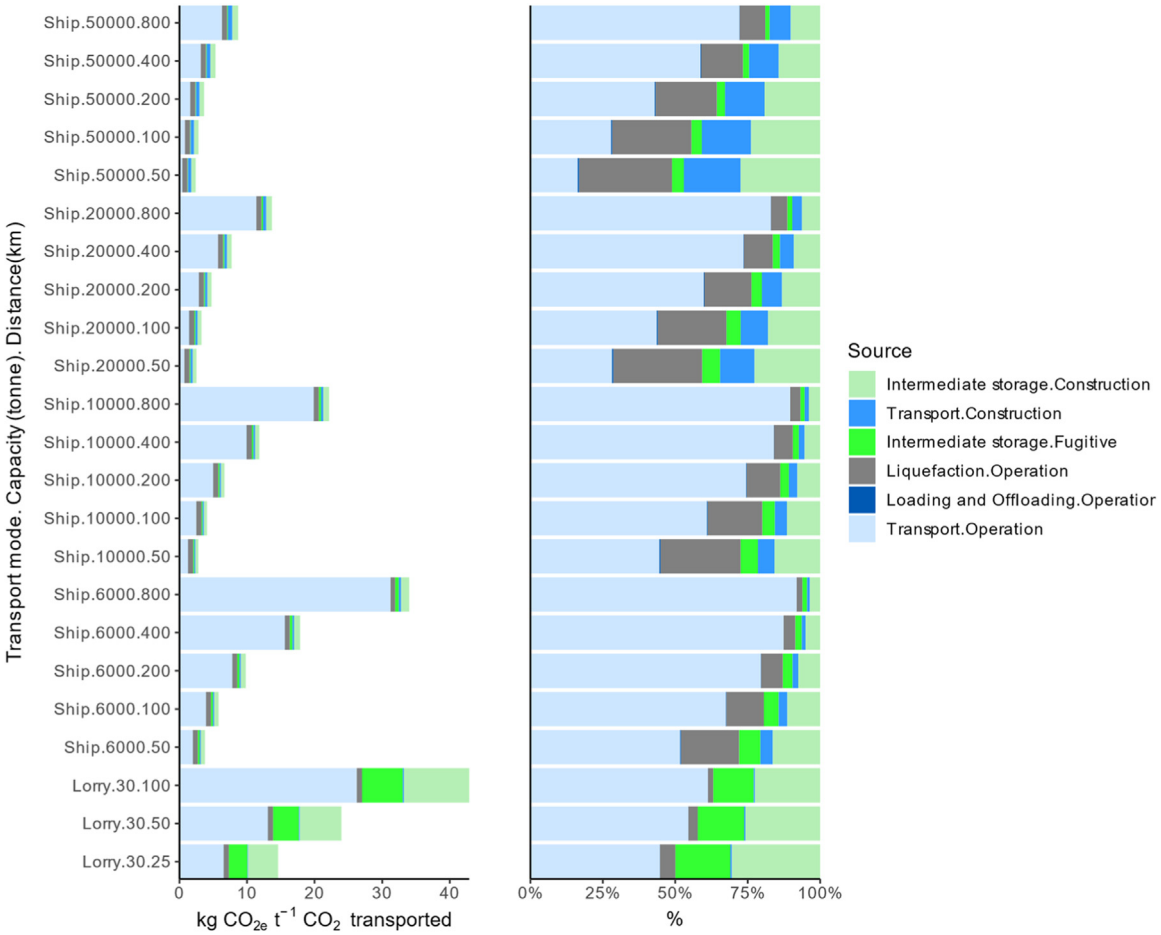


Fig. 11. GWP - CML 2016 midpoint absolute and relative results for batch-wise CO_2 transportation by ship and lorry for different distances and the 0.7 MPa pressure configuration.

Some assumptions must be made for the activity factor to allow the calculations. The first assumption is that the amount of CO_2 transported is equivalent to one cargo of the batch-wise transport mode considered. This batch is transported to an intermediate storage facility, considering a determined fixed distance for 30 years and performing the maximum number of trips possible. From those activity parameters, it is possible to calculate the GWP from twenty-three options from batch-wise transportation. From the ship options for 0.7 MPa pressure design, it was found that the greater capacity of the ship reduces its GWP, and the longer the distance travelled, the higher the relevance of the operation of the ship in total GWP. Overall, the operation of the ship shares higher than 50 % of the GWP indicator score in most cases, followed by the construction of the intermediate facility (i.e., storage tanks with intermediate storage hub). Liquefaction plays a more important role in the higher ship transport capacities. It is also possible to observe overall scale gains from higher ship capacities for all distances assessed. For lorry transportation, its operation exceeds 60 % of the total GWP, and the distance plays a significant role in the change of GWP.

Fig. 11 consolidates the absolute and relative GWP for all twenty-three options assessed.

The same generic calculations are performed for the 1.5 MPa pressure configuration for a transport ship with 10 kt of CO_2 storage capacity and are summarised in Fig. 12. The same pattern with the dominance of the operational costs on the emissions is observed for this pressure configuration.

Furthermore, a sensitivity analysis has been performed on the emission factors for the batch-wise fuel consumption and liquefaction energy consumption to assess the importance of likely variability for the overall GWP impact indicator score. The sensitivity range used for testing was ± 10 % step-wise change until cumulative ± 30 % variation is reached. The sensitivity is performed in ceterius paribus form, with no cross-impact accounted.

Fig. 13 illustrates the results of the sensitivity analysis in all twenty scenarios. The increase of impact on greater distances is observed across all twenty-eight scenarios, with the highest capacities batch-wise transportation options demonstrating less variation due to fewer trips from scale gains.

Multi-objective MILP cost and environmental optimisation for Northern Lights and Stella Maris CCS chains

As a final step in this paper the bottom-up LCI model developed for batch-wise transportation of CO_2 is used in a multi-period environment, developed as 5 periods of six years in each period, together with the cost minimisation MILP models developed in Fraga et al. (2021a, 2021b), for CO_2 transport and storage. To account for the GHG emissions of each chain element the elements from the chain (i.e., pipeline, well construction) are sourced from the literature and are detailed in the appendix section. This approach enables the model to characterise all elements temporally and establish how the optimal CCS network evolution solution may be affected depending on the emphasis placed on the emission minimisation objective as compared to cost.

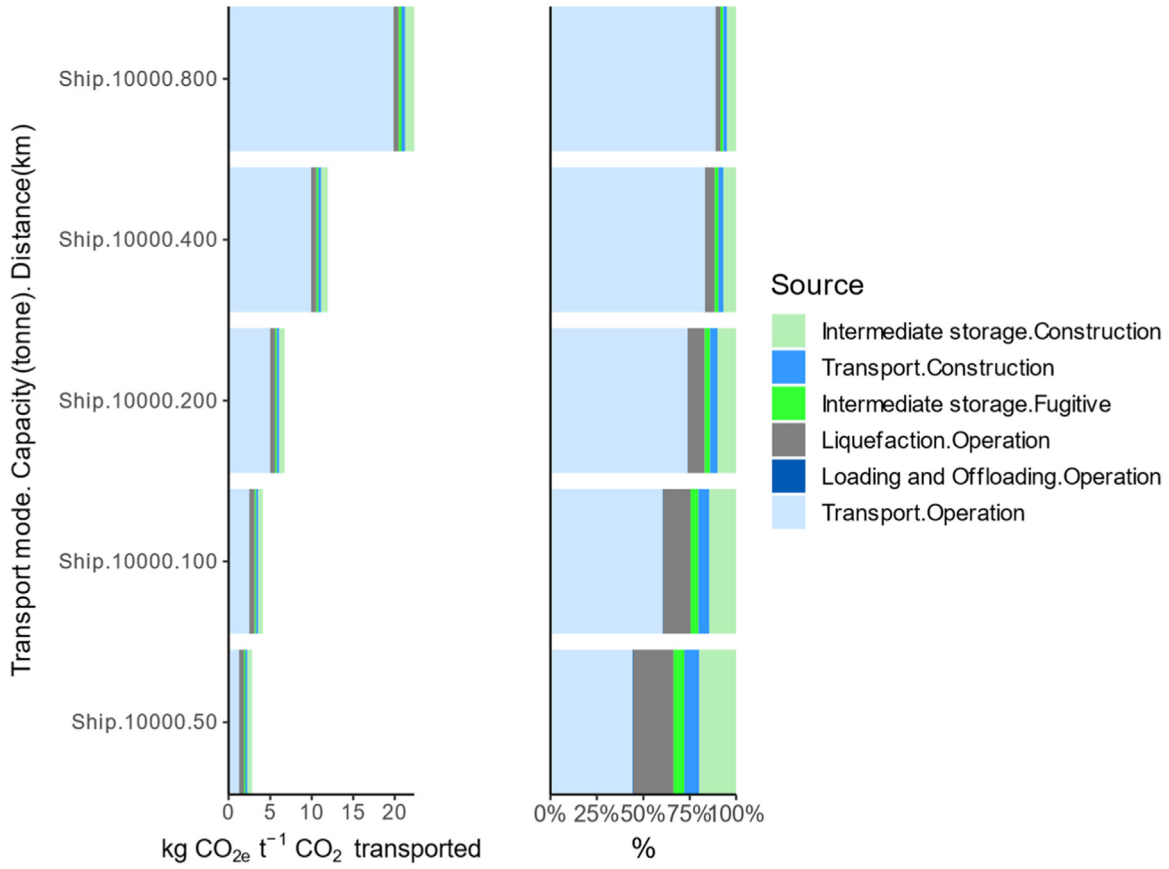


Fig. 12. GWP - CML 2016 midpoint absolute and relative results for batch-wise transportation of CO_2 by ship for different distances and for the 1.5 MPa pressure configuration.

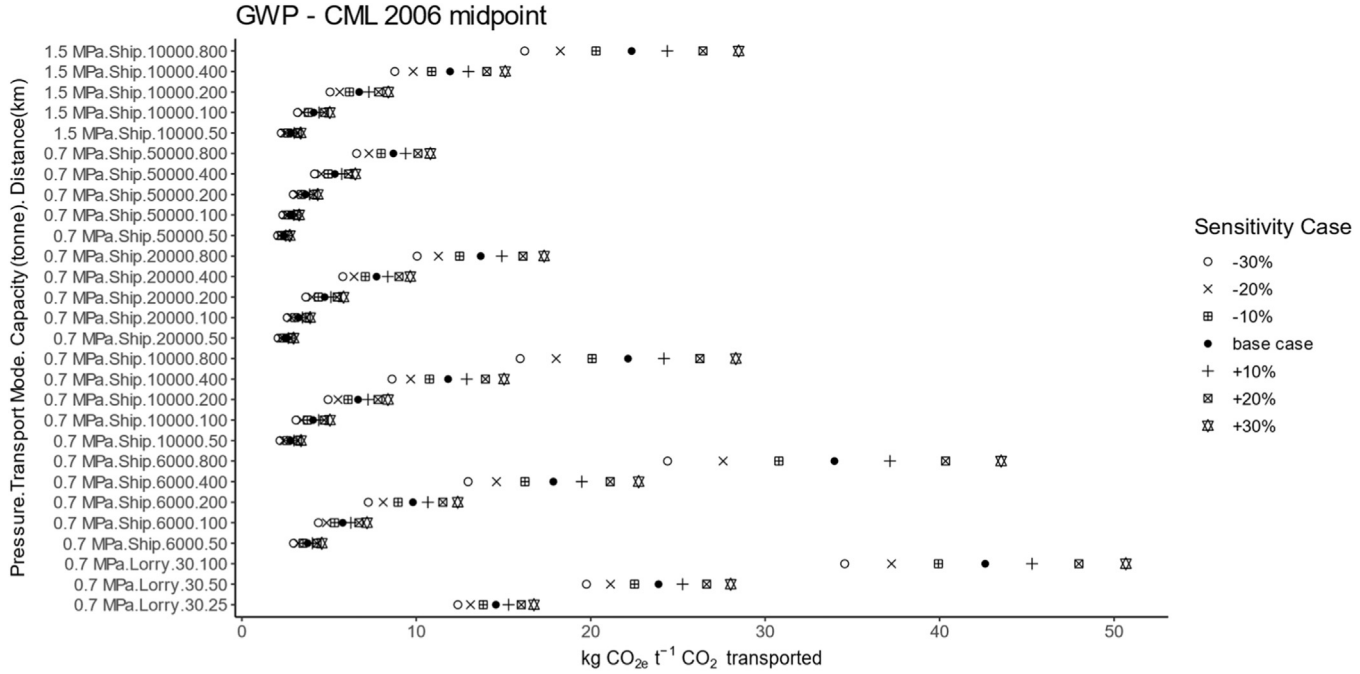


Fig. 13. Sensitivity analysis results for GWP - CML 2016 midpoint emissions to air from batch-wise transportation elements by ships in a distance range for both pressure configurations.

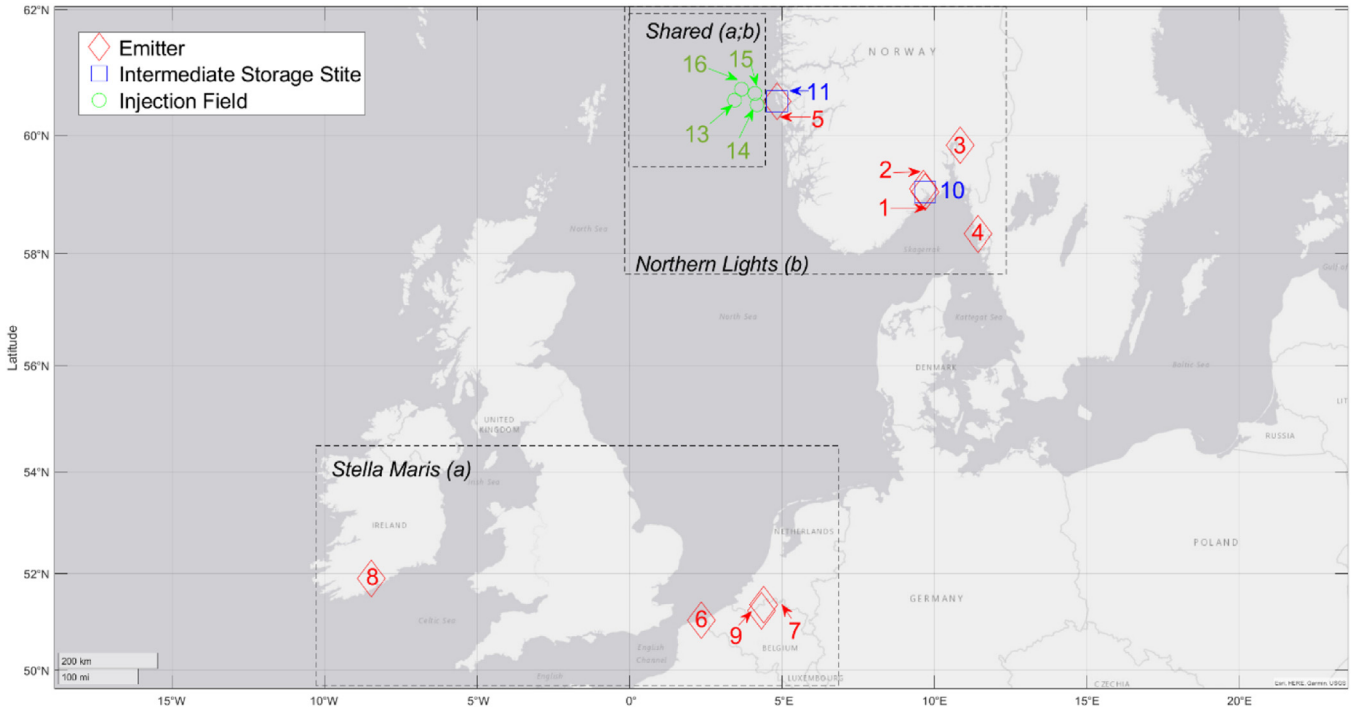


Fig. 14. Overview of CO_2 sources (1–9), intermediate storage site hubs (10,11), and injection fields (12–15) considered in this study – Yara (1), Norcem (2), Fortum Oslo Varme FOV (3), Preem (4), Kollsnes (5), Dunkirk (6), Antwerp (7), Ervia (8), SL-F1 (9), Hub-Grenland (10), Hub-Kollsnes (11), Aurora field (13) Smeaheia γ field (14), Smeaheia α (15), and Troll (16).

Northern Lights and Stella Maris CCS chains characterisation

The proposed model is implemented individually and then a third implementation has been devised by combining the two concepts. The first concept is an analogue of the first phase of the Northern Lights project which is already under-development. It comprises the transport and storage elements of the Norwegian Longship project, which covers the whole CCS including capture. It was recently appointed as a Project of Common Interest in Europe due to its cross-border ship transport infrastructure approach (Northern Lights, 2021). This CCS chain is ship dominated for collecting and transporting CO_2 to the intermediate storage site. Then, on from the intermediate storage, injection of CO_2 is intended to occur via an offshore pipeline to subsea wells and templates, tied back to operating platforms for power supply and umbilical connections. The second concept is the Stella Maris project, which only transports and stores CO_2 via direct injection ships (Altera Infrastructure, 2020). This concept is currently undergoing feasibility study assessment.

Fig. 14 illustrates all elements that comprise both concepts. This implementation assumes that the geological storage sites for the Stella Maris Concept are the same as the ones planned for the Northern Lights project, which are located at the Horda platform offshore Norway.

The Northern Lights concept considers CO_2 captured in four sites. The first two sites are in Grenland, including Yara (1), a fertiliser plant at Herøya in Porsgrunn, and Norcem (2), a cement plant in Brevik. The third is Fortum Oslo Varme -FOV (3), a waste incineration plant in Klemetsrud, near Oslo. The fourth is Preem (4), a refinery in Lysekil, Sweden. This concept uses two intermediate storage Hubs, Hub-Grenland (10) in Grenland, and Hub-Kollsnes (11), to establish a transport route of CO_2 by ship towards the storage sites Aurora (13), Smeaheia γ (14), Smeaheia α (15), and Troll (16), which are connected to Hub-Kollsnes (11), via offshore pipeline (Fraga et al., 2021a).

The Stella Maris Concept contains three ports concentrating the CO_2 captured in the surrounding industry clusters. In France, the Dunkirk (6) port is considering the evolution of the 3D project, which will start its deployment at ArcelorMittal's site in 2025, with increasing CO_2 supply from 1 Mtpa to 10 Mtpa (3D Project, 2022). In Belgium, the Antwerp port is a source of CO_2 captured from multiple chemical processing plants (Global CCS Institute, 2021b). This port also has SL-F1 (9), a hypothetical small-scale emitter intended to demonstrate how the model can incorporate small-scale CO_2 transported by lorries. The Ervia (8) port, located in Cork, Ireland, concentrates the CO_2 collected from the Cork industrial cluster, which has had a Memorandum of Understandings (MOU) signed with Equinor to study the potential of CCS in their power generation and refining sectors (Global CCS Institute, 2021b; Ervia, 2019).

Table 14
CO_2 Initial storage capacity parameters.

Injection field	Available from period	Initial storage capacity (Mt)
Aurora (13)	1	206
Smeaheia γ (14)	2	1,800
Smeaheia α (15)	2	45
Troll (16)	5	4,000
Total		**6,051**

The emission profile ($EP_{e,t}$) for each emitter is shown in Fig. 15. The Northern Lights Concept (1–5) starts with 1.1 Mtpa in the first period and reaches 8.3 Mtpa in the last period, meaning an CAGR of 7 %. The Stella Maris Concept chain (6–9) starts with 0.3 Mtpa in the first period and reaches 15.1 Mtpa in the last period, equivalent to an CAGR of 14 %. The combined concept starts with 1.4 Mtpa and reaches 23.4 Mtpa in the last period with an CAGR of 10 %. The total amount of CO_2 captured for the combined concept during the 5 periods (i.e.,30 years) is 296.3 Mt.

The captured CO_2 is considered to be injected into four distinct geological storage sites with different availability and totalling 6051 Mt of initial storage capacity, as represented in Table 14. The initially studied file for injection was Smeaheia γ (13) and Smeaheia α (14). As the development concept has matured in recent years, it has changed to start with the Aurora (12) field in the Johansen formation. While Troll (15)

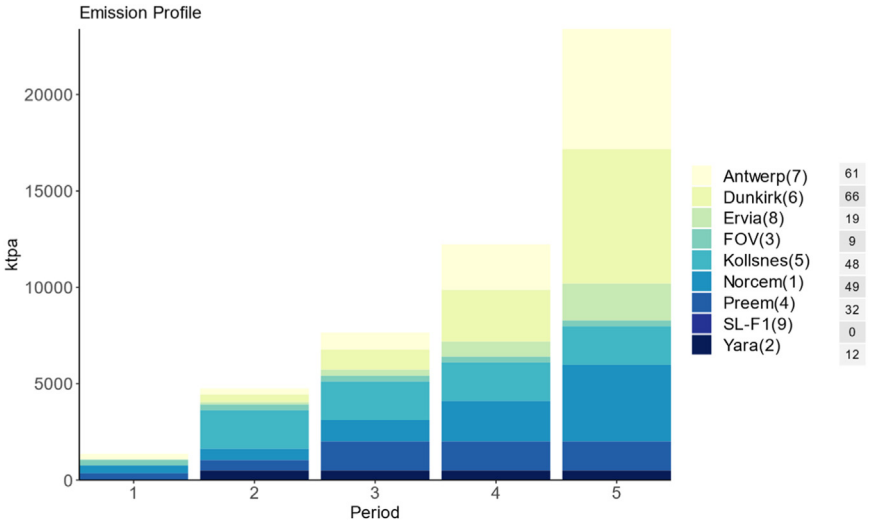


Fig. 15. Emission profile considered in the MILP model implementation in ktpa. The total emission is 296 Mt CO_2. The total emitted CO_2 per source over the time horizon (i.e., in Mt) is noted in the grey shaded text.

Table 15
CO_2 stream purity parameters (Equinor, 2020).

Compound	Value ppm (mol)
H_2O	≤30
NOx	≤10
SOx	≤10
CO	≤100
O_2	≤10
H_2S	≤9
H_2	≤50
Hg	≤0.03

is still a significant gas producing field, its availability as a storage field has been considered to occur in the last period of the chain.

The CO_2 entering the network is assumed to have been purified, and its specification to be within the parameters shown in Table 15 in line with Equinor (2020). Moreover, all the cost figures are updated to 2022 using the harmonised index for consumer price and exchange rates from Eurostat (2022).

Multi-objective MILP cost and environmental optimisation model formulation

The objective function chosen aims to minimise the CCS chain total cost ($TC_{(k,i,j,t)}$) with a weight w_1 and the total emissions ($TE_{(k,i,j,t)}$) with weight w_2. The weights are used to help explore the importance of the relative weight chosen in the optimal solution identification. The sum of weights equals to 100 %.

$$\min w_1 \sum_{(k,i,j,t)} TC_{(k,i,j,t)} + w_2 \sum_{(k,i,j,t)} TE_{(k,i,j,t)} \tag{2}$$

The $TC_{(k,i,j,t)}$ formulation is formed by CAPEX and OPEX for conditioning, transport, intermediate storage and geological storage as defined in Fraga et al. (2021a, 2021b). The total emissions ($TE_{(k,i,j,t)}$) are described in detail in this section and equal the sum of total transportation emissions ($TTE_{(k,i,j,t)}$), conditioning emissions ($TCE_{(k,i,j,t)}$), intermediate storage emissions ($TIE_{(k,i,j,t)}$) and geological storage emissions ($TSE_{(f,t)}$) as defined in Eq. (3). Each emission element results from combining the proposed gate-to-gate LCI model developed in this research and existing literature values for other elements. This model is subject to the same constraints of mass balance, transport and injection wells assignment for the cost minimisation presented in Fraga et al. (2021a, 2021b).

$$TE_{(k,i,j,t)} = TTE_{(k,i,j,t)} + TCE_{(k,i,j,t)} + TIE_{(k,i,j,t)} + TSE_{(f,t)} \tag{3}$$

The emission models use a generic application to estimate emissions of GHG from the chain, considering an emission factor element and an activity factor element, as described in Eq. (4).

$$Mass\ CO_2e\ emitted = Emission\ factor \times \left(\frac{t\ CO_2e}{activity}\right) \times \text{Activity } factor \tag{4}$$

Table 16 contains all the notations considered in the emissions model and is further explained. The sets and subsets are equal to the cost optimisation model.

The GHG emissions are estimated as a GWP impact category score for 100 years. The transport element emissions ($TTE_{(k,i,j,t)}$) are formed by constructing ($TTE_{(k,i,j,t)_{cons}}$) and operating ($TTE_{(k,i,j,t)_{op}}$) the batch-wise element (i.e., ships and lorries) and pipelines. Eq. (5) details the composition of the total emission from the transport of CO_2.

$$TTE_{(k,i,j,t)} = TTE_{(k,i,j,t)cons} + TTE_{(k,i,j,t)op} \tag{5}$$

The construction emissions from the transport element considers the evolution of the infrastructure along the time horizon with the individual emission factor from each transport option. Eq. (6) presents this relationship.

$$TTE_{(k,i,j,t)_{cons}} = \sum_t (NS_{(s,t)} - NS_{(s,t-1)})\ \mathrm{EF}_{s\,cons} + (NU_{(p,i,j,t)} - NU_{(p,i,j,t-1)})W_{p,i,j}\mathrm{EF}_{p\,cons} + \left(NL_{(l,t)} - NL_{(l,t-1)}\right)\mathrm{EF}_{p\,cons} \forall s, l, p \in k \tag{6}$$

Where:

$NS_{(s,i,j,t)}$ = Number of ships s required in period t

$NU_{(p,i,j,t)}$ = Number of units required from transport mode p, between locations i and j, in period t

$NL_{(l,i,j,t)}$ = Number of lorries l required in period t

EF_{scons} = ship s construction factor (t CO_2e ship^{-1})

W_{ij}= weight of the pipeline p between location i and j (t)

EF_{Pcons} = Pipeline p construction factor (t CO_2e t^{-1} pipeline)

EF_{scons}= Lorrie l construction factor (t CO_2e lorry^{-1})

The operation-related emission from the transport element results from fuel combustion for batch-wise transportation and compression (i.e., power for booster stations) for offshore pipeline transportation options. Fugitive emissions are accounted for in all transport modes. Eqs. (7) to (10) detail the operational emissions from each transport mode explored in this research.

$$TTE_{op} = TTE_{sop} + TTE_{pop} + TTE_{lop} \forall\ s, l, p \in k \tag{7}$$

$$TTE_{sop} = \sum_t (NU_{(s,i,j,t)} \frac{TRIP_{(s,i,j)}}{24} EF_{sop1} + NU_{(s,i,j,t)} EF_{sop2}$$

Table 16
MoMILP model sets, subsets, parameters, and variables for emissions model.

Continuous Non-Negative Variables	
$F_{(k,i,j,t)}$	Annual flow rate (tpa) of CO_2 from transport mode k, between locations i and j, in period t
$IF_{(f,t)}$	Injected (t CO_2) at field f, in period t
$RC_{(f,t)}$	Remaining injection capacity (t CO_2) at field f, in period t
$NS_{(s,i,j,t)}$	Number of ships s required from location i to location j in period t
$TTE_{(k,i,j,t)}$	Total Transport Emissions (t CO_2e) from transport mode k, between locations i and j, in period t
$TCE_{(k,i,j,t)}$	Total Conditioning Emissions (t CO_2e) from transport mode k, between locations i and j, in period t
$TIE_{(k,i,j,t)}$	Total Intermediate Emissions (t CO_2e) from transport mode k, between locations i and j, in period t
$TSE_{(f,t)}$	Total Geological Storage Emissions (t CO_2e) from field f, in period t
$TE_{(k,i,j,t)}$	Total Emissions (t CO_2e) from transport mode k, between locations i and j, in period t
Integer Variables	
$NU_{(k,i,j,t)}$	Number of units required from transport mode k, between locations i and j, period t
$NS_{(s,t)}$	Number of ships s required in period t
$NL_{(l,t)}$	Number of lorries l required in period t
$NIU_{(f,t)}$	Number of integration units at field f and period t
$NAW_{(f,t)}$	Number of additional wells at field f and period t
Binary Variables	
$AN_{(k,i,j,t)}$	1, if a node i is activated from transport mode k, between locations i and j, in period t
$ACH_{(i,t)}$	1, if a node i is activated from compression and heating of CO_2, in period t
$AI_{(i,t)}$	1, if a node i is activated for intermediate storage
$AIH_{(i,t)}$	1, if a node i is activated for intermediate storage hub

$$+ NU_{(s,i,j,t)} \frac{TRIP_{(s,i,j)}}{EF_{sop3}} EF_{sop4} TC_s)\, YP_t \forall\, s \in k \quad (8)$$

$$TTE_{pop} = \sum_t (NU_{(p,i,j,t)}\, d_{pij}\, EF_{pop1} + P_{(p,i,j)}\, 8760 \times EF_{pop2} + P_{(p,i,j)} EF_{pop3}) YP_t \;\forall\, p \in k \quad (9)$$

$$TTE_{lop} \sum_t NU_{(l,i,j,t)}\, 2d_{lij}\, EF_{lop1} YP_t \;\forall\, l \in k \quad (10)$$

Where:

$NU_{(k,i,j,t)}$ = number of units required from transport mode k, between locations i and j, in period t

$TRIP_{(s,i,j)}$ = total time for a round trip (days) with transport ship s between i and j

EF_{sop1} = ship s operation emission factor for fuel combustion (t CO_2e day^{-1})

EF_{sop2} = ship s operation emission factor for loading and unloading (t CO_2e $trip^{-1}$)

YP_t = years within a period t

EF_{sop3} = ship s pressure build-up allowance (days)

EF_{sop4} = ship s boil of rate (percentage)

TC_s = maximum transport capacity (t) for ship s

d_{pij} = pipeline p haversine distance (km) between i and j

EF_{pop1} = fugitive emission factor for pipeline p (t CO_2e km^{-1})

$P_{(p,i,j)}$ = power required for a booster to overcome the pressure drop of pipeline p (MW), between locations i and j

EF_{pop2} = electricity emission factor for a booster in pipeline p (t CO_2e MWh^{-1})

EF_{pop3} = fugitive emission factor for a booster in pipeline p (t CO_2e MW yr^{-1})

d_{pij} = lorry l route distance (km) between i and j

$EF_{l_{op2}}$ = lorry l operation emission factor for fuel combustion (t CO_2e km^{-1})

The emission is calculated based on the operational phase for the conditioning element of the CCS chain. For liquefaction and compression, the electricity required to liquefy CO_2 is the emission source. In addition, fugitive emissions are considered. Eq. (11) details the emissions from the conditioning element of the CCS chain.

$$\text{TCE} = \sum_t F_{(k,i,j,t)}\, EF_{cop1} YP_t \forall\, s,p \in k;\; \forall e,h \in i \quad (11)$$

Where:

$F_{(k,i,j,t)}$ = annual flow rate of CO_2 from pipeline p, emitter e and intermediate storage h, and from ship s from locations e and h to location j, in period t

EF_{cop1}= electricity emission factor for liquefaction of CO_2 (t CO_2e t^{-1} CO_2)

YP_t = years within a period t

The intermediate storage element of the chain is estimated by the construction of intermediate storage tanks and sites either in stationary emitters of intermediate storage hubs, the administrative facility is considered only for intermediate storage hubs, and fugitive emissions are considered for both stationary emitters and intermediate storage hub, as detailed in Eq. (12).

$$\begin{aligned}\text{TIE} = \sum_t &\big((NS_{(s,t)-}\; NS_{(s,t-1)}) + (NL_{(l,t)-}\; NL_{(l,t-1)})\text{EF}_{icons(k)} \\ &+ (AIH_{(i,t)} - AIH_{(i,t-1)})\text{EF}_{icons2} \\ &+ (AI_{(i,t)} - AI_{(i,t-1)})\text{EF}_{icons3} 8760\big) YP_t \;\forall\, s,l \in k\end{aligned} \quad (12)$$

Where:

$NS_{(s,i,j,t)}$ = number of ships s required in period t

$NL_{(l,i,j,t)}$ = number of lorries l required in period t

$\text{EF}_{icons(k)}$ = emission factor for intermediate storage tank and site construction for transport mode k (t CO_2e t^{-1} CO_2 stored)

$AIH(i,t) = 1$ if the intermediate storage hub is activated in the location i and period t; 0 otherwise

EF_{icons2} = emission factor for intermediate storage management building construction (t CO_2e $facility^{-1}$)

$AI(i,t) = 1$ if intermediate storage facility is activated in the location i and period t; 0 otherwise

$\text{EF}_{c\,op2}$ = fugitive emission factor for intermediate storage facility and (t CO_2e hr^{-1})

YP_t = years within a period t

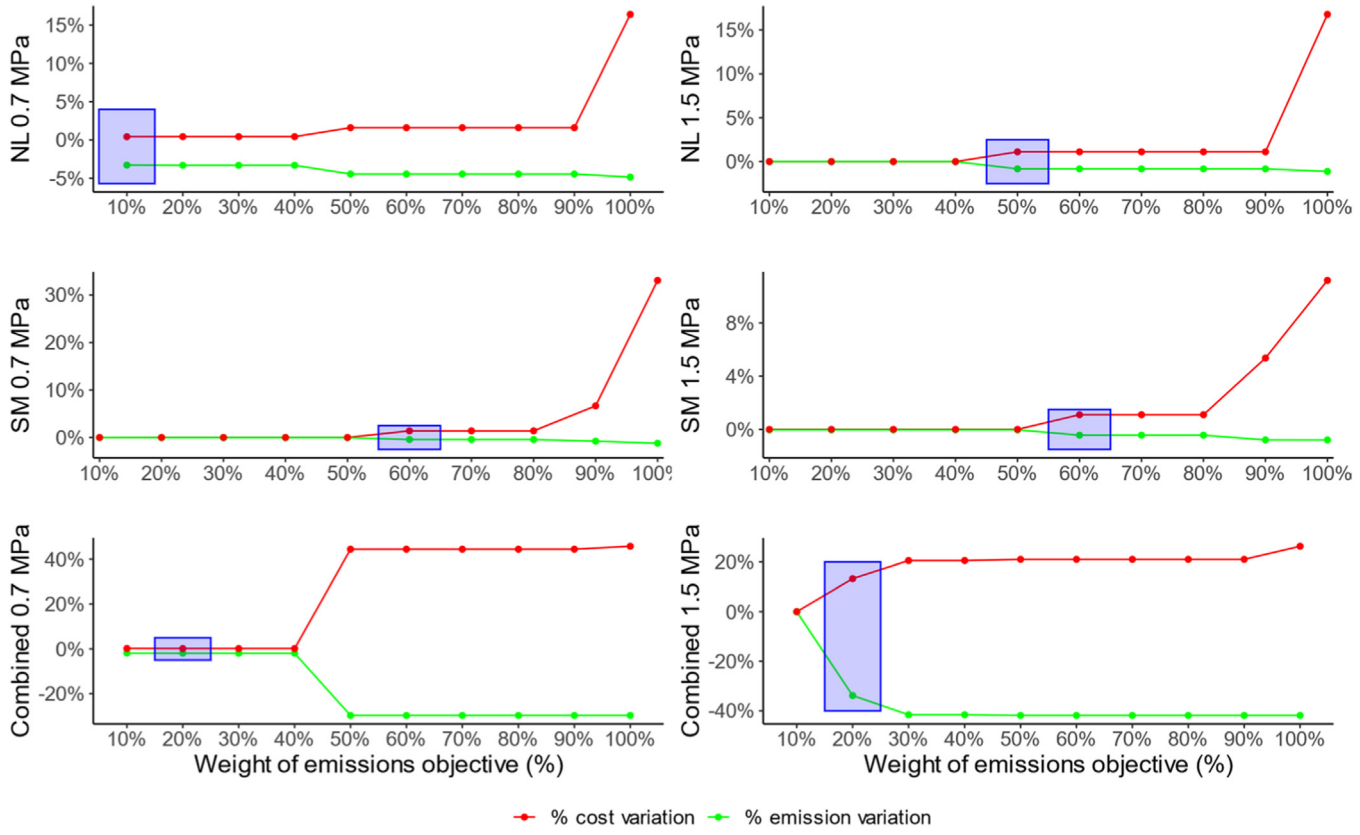


Fig. 16. Total cost and total emissions variations from MoMILP emission objective function weight in all concepts and both pressure configurations.

The last element of the emission model is the injection wells emissions, considering the construction and its fugitive emissions, as described in the Eq. (13).

$$\begin{aligned}\mathrm{TSE}_{(f,t)} = &\left(NIU_{(f,t)} - NIU_{(f,t-1)} + NAW_{(f,t)} - NAW_{(f,t-1)}\right)EF_{s_{cons}} \\ &+ \left(\left(NIU_{(f,t)} + NAW_{(f,t)}\right)EF_{s2}8760\right)YP_t \forall\ f \in i \end{aligned} \quad (13)$$

$NIU_{(f,t)}$ = integration unit at storage field f and period t
$NAW_{(f,t)}$ = satellite well storage field f and period t
$\mathrm{EF}_{scons(k)}$ = emission factor well construction (t CO_2e $well^{-1}$)
EF_{s2} = fugitive emission factor per construction (t CO_2e $hour^{-1}$)
YP_t = years within a period t

Results and discussion

Fig. 16 summarises the MoMILP solutions of all concepts and pressure configurations, highlighting the best solution, defined as the cost increase with the highest emission reduction. It is observed that the pressure configuration of the concept did not change the best solution, meaning all concepts have their best at the same point. It is also noted that the highest weight of the emission minimisation resulted in a substantial cost increase without an equivalent emission minimisation (i.e., 100 % solution).

The best solution for the NL concept implementation at 0.7 MPa pressure configuration was found at 10 % emissions weight, resulting in a 0.4 % total cost increase and −3.3 % total emission reduction. For the 1.5 MPa pressure configuration, the best solution was found at 50 % emission weight, resulting in 1.1 % and −0.8 % cost increase and total emission reductions, respectively.

The best solutions in the SM concept for both pressure configurations were found at 60 % emissions weight, resulting in 1.4 %; −0.4 %, and 1.1 %; −0.4 % variations for total cost and emissions, respectively, for 0.7 and 1.5 MPa pressure configurations.

Lastly, the Combined concept best solution for 0.7 MPa was found at 20 % emission weight for both pressure configurations. For 0.7 MPa, variations of 0.2 %; and −1.9 % for total cost and emissions were obtained. At 1.5 MPa, the best solution resulted in 13.3 %; and −33.8 % cost increase and emission reduction, respectively.

In addition to the impact of the increase on the emissions objective function in the multi-objective solutions, Fig. 17 consolidates the derived pareto front from each solution of the step-wise implementation for all concepts with label points where there was a change in the network.

In the NL concept, the best solution (i.e., points a.1 and c.2 in Fig. 17) resulted in 32.8 and 46.3 kg CO_2e t^{-1} CO_2 stored and 4929.7 and 6958.7 kt CO_2e emissions through the time horizon, respectively, for both pressure configurations. Fig. 18 demonstrates the breakdown of the overall emissions per emission category and as a functional unit through the time horizon for both pressure configurations. The first observation is the dominance of the batch-wise transportation operation on the overall emissions and through the time horizon for both pressure scenarios. Also, with the setup of the CCS infrastructure in the first period in each concept configuration, it can be observed that the unit emission decreases through time as the emission profile increases and the need for additional infrastructure is reduced. In the 0.7 MPa pressure configuration, the best solution added a mid-capacity transport ship (i.e., 20 kt and in point a.1 in Fig. 17) that would result in fewer trips at later periods. An additional increase in the emission weight (i.e., points a.1 to f.1 in Fig. 17) indicated the avoidance of constructing a satellite well in Smeaheia γ (14) by increasing injection rates in the Aurora (13) field. Since the 1.5 MPa ship capacity is constrained, the best solution

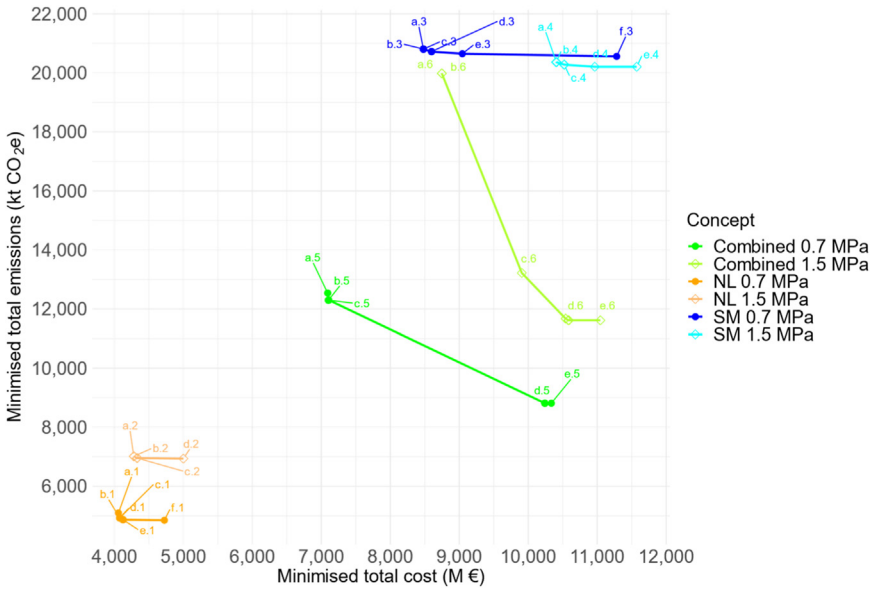


Fig. 17. Total cost and total emissions for all concepts and both pressure configurations. Points (i.e., from a to f) are illustrated, indicating there is a modification in the network.

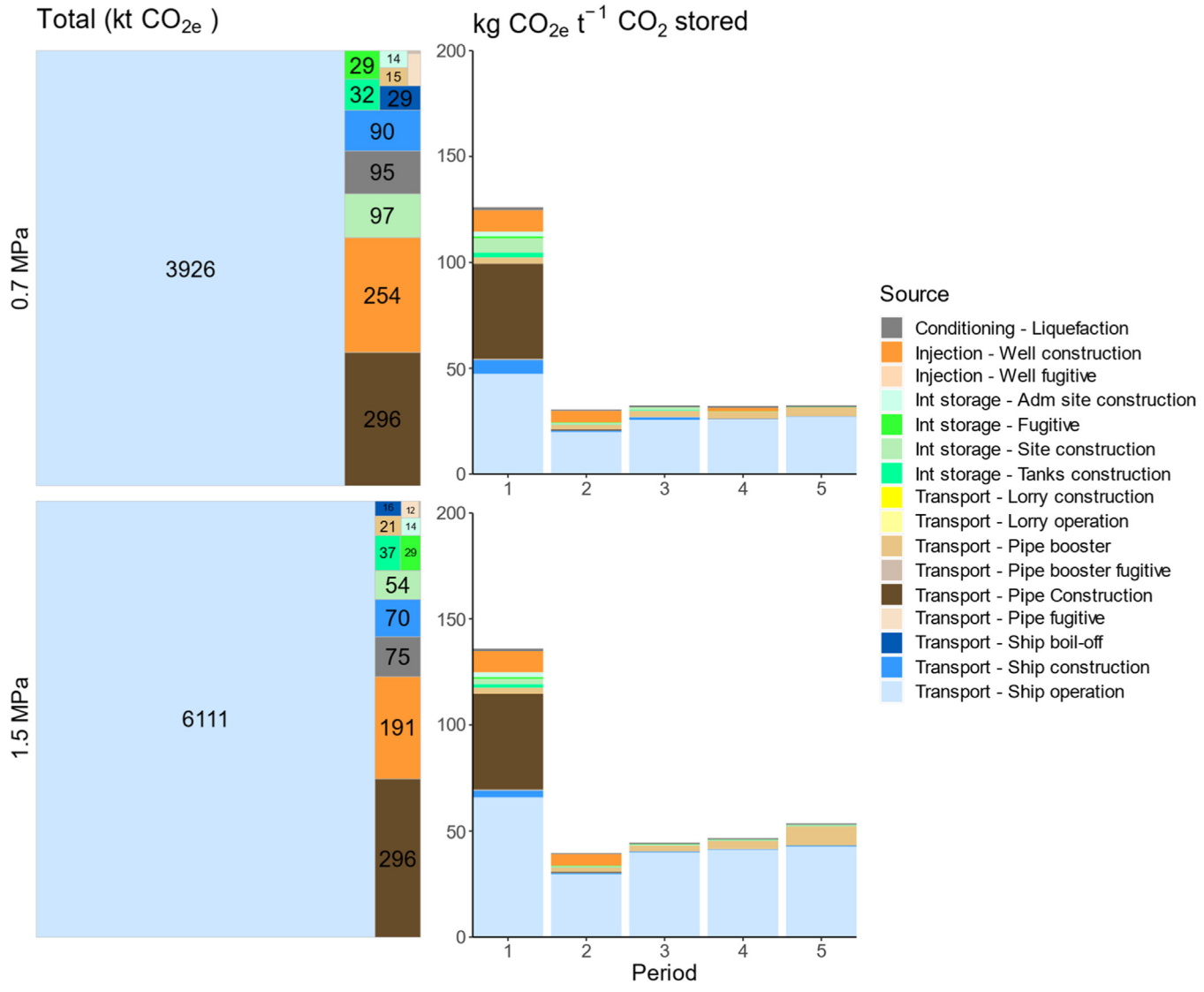


Fig. 18. NL concept best solutions absolute GHG emissions and results per chain element for the 5 periods represented with the reference unit.

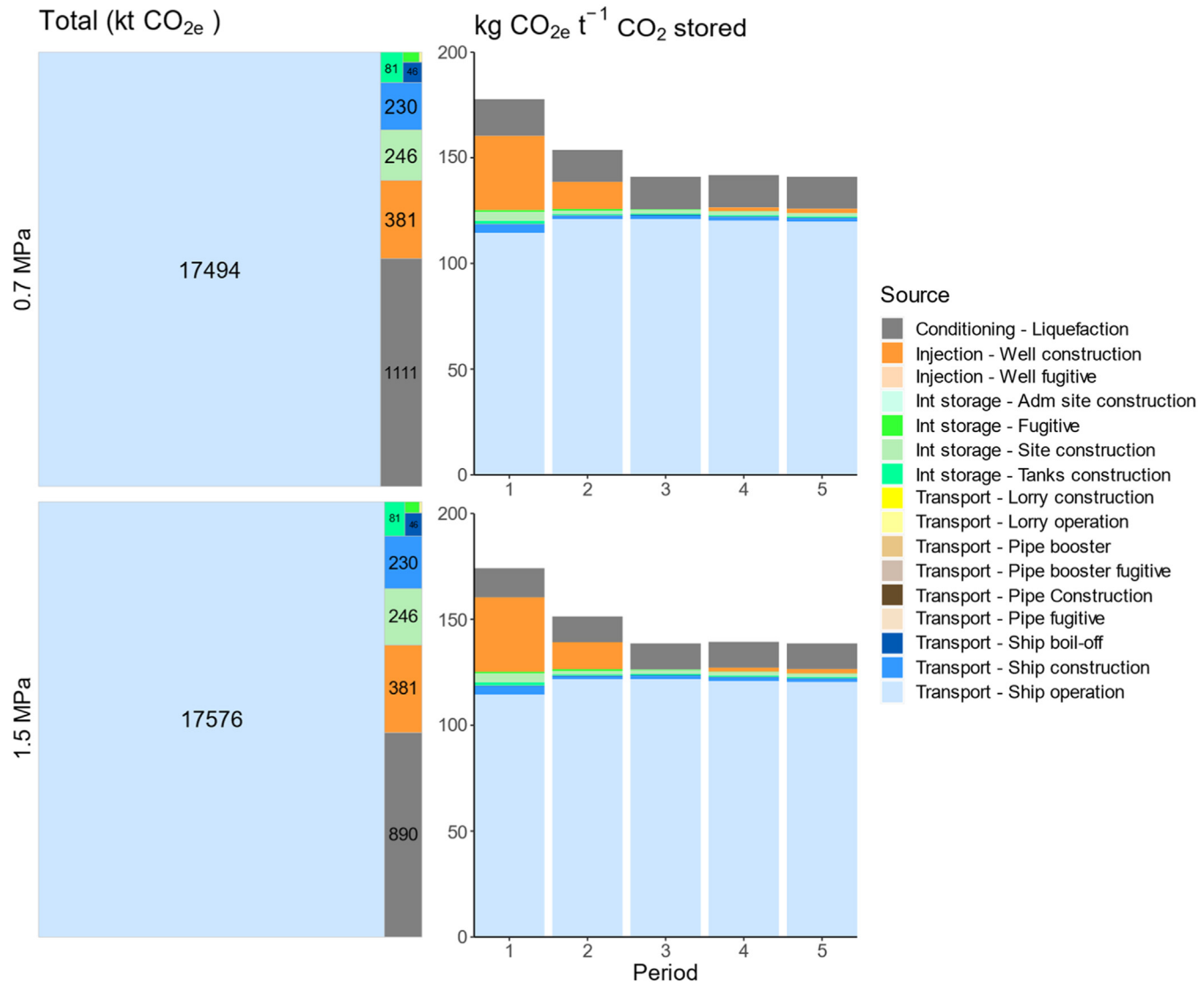


Fig. 19. SM concept best solutions absolute GHG emissions and results per chain element for the 5 periods represented with the reference unit.

(i.e., point c.2 in Fig. 17) avoided the construction of an additional well, while increasing the injection rate at a more expensive geological storage site (i.e., in Aurora (13) field). Moreover, the 100 % emission weight solution (i.e., points e.3 to f.3 and d.4 to e.4 in Fig. 17) concentrates all the injection of CO_2 in higher cost geological storage site (i.e., Aurora (13)).

The SM concept best solution (i.e., points d.3 and c.4 in Fig. 17) resulted in 142.1 and 139.0 kg CO_2e t^{-1} CO_2 stored and 20,721.7 and 20,279.6 kt CO_2e emissions through the time horizon, respectively, for both pressure configurations (i.e., 0.7 MPa and 1.5 MPa). The resulting pattern is similar in both pressure configurations as this concept is dominated by the direct injection ships with one capacity (i.e., 10 kt capacity). The changes on the network are also shared between both pressure configurations and indicate higher injection rates in a more expensive field (i.e., Aurora (13), up to points c.5 and b.6 in Fig. 17) by making shorter ship trips from Ervia (8), thus resulting in fewer emissions from ship operations. The higher the emission weight in the objective function, the more ship routes are established for injection in Aurora (13)) (i.e., from Dunkirk and Antwerp), resulting in reduced emissions (i.e., points c.5 and b.6 in Fig. 17 but to a less degree than the best solution (i.e., at 60 % emission weight). Fig. 19 presents the total emissions breakdown and reference unit results per period for the best solution.

The Combined concept best solution (i.e., points c.5 and c.6 in Fig. 17) at 20 % emission weight resulted in 41.5 and 39.2 kg CO_2e t^{-1} CO_2 stored and 12,307.8 and 11,617.7 kt CO_2e emissions through the time horizon, respectively, for both pressure configurations (i.e., 0.7 MPa and 1.5 MPa). In this implementation, although the best solution was found at the same emission weight for both pressure configurations, the changes on the network were distinct. In 0.7 MPa configuration, the best solution produced a modification on the transport ship fleet design, adding intermediate capacity ships (i.e., 20 kt) to operate in the chain. For 1.5 MPa pressure configuration, the modification on the network consists in replacing some of the small-capacity transport ships (i.e., 10 kt capacity) from Antwerp (7) to Hub-Kollsnes (12), limiting this ship route to only 32 trips per year, resulting in thirteen fewer ships operating on the chain (i.e., maximum fleet of fourteen transport ships). The pipeline installation occurs in the third period when the CO_2 flow rate is the highest. For further emission weight solutions (i.e., higher than 20 %), the same modification occurs with Dunkirk (6) being con-

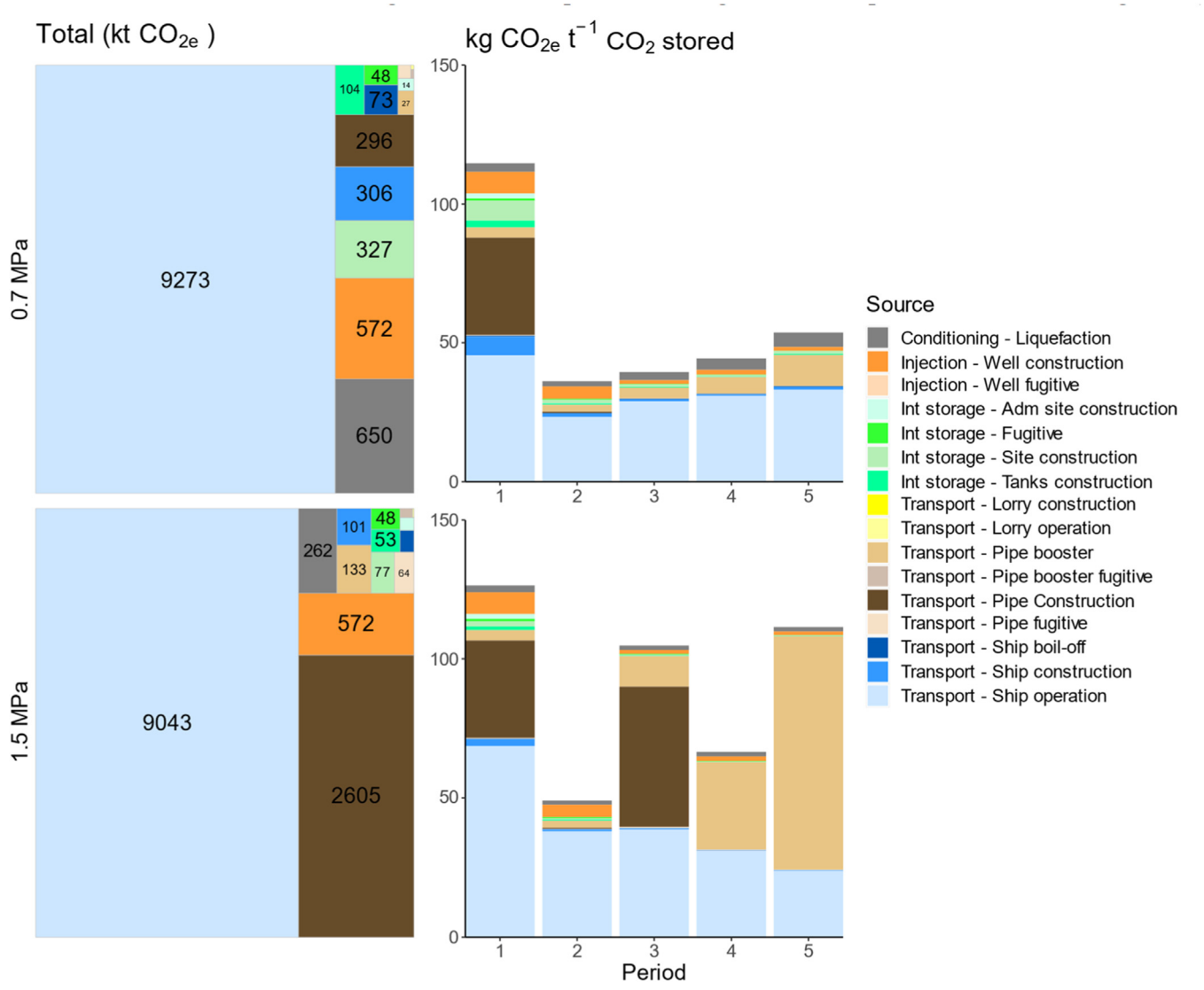


Fig. 20. Combined concept best solutions absolute GHG emissions and results per chain element for the 5 periods represented with the reference unit.

nected to Antwerp (7) via offshore pipeline (i.e., instead of transport ship), along with the injection in higher cost geological storage site (i.e., Aurora (13)) until the time that one additional well is not required (i.e., points d.6 and e.6 in Fig. 17). That pattern (i.e., replacement of ship route by offshore pipeline) is also observed in solutions from 50 % emission weight for 0.7 MPa pressure configuration (i.e., points d.5 and e.5 in Fig. 17).

Fig. 20 summarises the breakdown for total emissions and results per reference unit over time. Overall, the ship operations-related emissions are dominant in both pressure configurations. The reference-unit emission through the time horizon is unequal for both pressure configurations. The lower pressure reflects the installation of infrastructure in the first period, decreasing in the second period and increasing further later, as the CO_2 supply rates grow. For 1.5 MPa pressure configuration, the infrastructure is set in the first and third periods to accommodate the growing CO_2 supply profiles. The third period indicates the installation of a pipeline to connect Antwerp (7) to Hub-Kollsnes (12), with the booster's emissions dominating the reference unit as ship trips are limited.

Conclusions

MILP formulation of the emissions accounting that, combined with the deterministic cost minimisation, has been used to devise the multi-objective cost and emission minimisation model developed. The adopted approach involved the sum of attributed weight to both objective functions (i.e., cost and emissions of GHG) with their sum limited to 100 %. This model was implemented for the six CCS chain implementations. During this process, a 10 % step-wise increase to the weight of the emissions objective was applied. This incremental enhancement represented a methodical and deliberate escalation in the emission objective relevance within the two objectives. All solutions from the MoMILP implementation were kept and compared with the solution where the weight of the emissions objective function is 0 % (i.e., cost minimisation only). The best solution for each implementation was selected and defined as the one with the cost increase that resulted in the highest relative emission reduction.

The Northern lights (NL) concept best solution was found with a 10 % emission weight on the objective function and produced and cost

increase of 0.4 % to achieve an emission reduction of −3.3 %, totalling 4,929.7 kt CO_2e, 32.8 kg CO_2e t^{-1} CO_2 stored and a relative emission of 3.3 %, for 0.7 MPa pressure configuration. For the higher pressure configuration, the best solution was found with the emission objective function weight at 50 %, resulting in a cost increase of 1.1 % and emission reduction of −0.8 %, resulting in 6,958.7 kt CO_2e, 46.3 kg CO_2e t^{-1} CO_2 stored and a relative emission of 4.6 %. This implementation reduced emissions by avoiding the construction of additional injection infrastructure in the geological storage field with lower costs.

For the Stella Maris (SM) concept, the best solution was identified with a 60 % weight in the emission objective function for both pressure configurations, resulting in 20,721.7; and 20,279.6 kt CO_2e, 142.1; 139.0 kg CO_2e t^{-1} CO_2 stored with a relative emission of 14.2 % and 13.9 % for 0.7 and 1.5 MPa, respectively. The changes in the network are similar in both pressure configurations and relate to higher injection rates in a more expensive field (i.e., Aurora (13)), resulting in shorter ship trips from distal stationary emitters (i.e., Ervia (8)).

The combined concept best solution was found with 20 % and 20 % emission weight, resulting in 41.5 and 39.2 kg CO_2e t^{-1} CO_2 stored, 12,307.8 and 11,617.7 kt CO_2e, and relative emissions of 4.2 % and 3.9 % relative emissions, respectively, for both pressure configurations (i.e., 0.7 MPa and 1.5 MPa). In the 0.7 MPa configuration, the best solution was obtained by adding intermediate capacity ships (i.e., 20 kt) to operate in the chain, resulting in fewer trips. For the 1.5 MPa scenarios, the best solution consisted of replacing small-capacity transport ships (i.e., 10 kt capacity) for an offshore pipeline.

Most of the parameters considered in the models developed are project-, site- or region-specific; therefore, even though it is possible to infer qualitative conclusions from the six implementations performed, they should not be generalised to every CCS chain, although they can be used as starting hypotheses for the analysis of any particular CCS chain. The following qualitative conclusions were obtained from the implementations presented:

- When combining both concepts (i.e., NL and SM), the lower-cost concept prevailed over the network in both deterministic and stochastic implementations.
- The step-wise runs allowed to identify the trade-offs on the CCS network evolution, and although the best solution was case-specific, the same trade-offs occurred in all concepts, when possible, with higher emission weight in the objective function. The first modification was to add mid-sized transport ship capacity (i.e., 20 kt) to reduce the number of trips while marginally increasing the cost. This was followed by avoiding the emissions of installing an additional well in a cheaper geological storage site, thus increasing the injection costs while reducing emissions. Thirdly, reducing the emission from ship trips by injecting CO_2 into a closer and more expensive geological storage site was observed. The last and most drastic modification was that of replacing a ship route with a pipeline segment, thus substantially reducing the emissions (i.e., −33.8 %) while also increasing the total cost substantially (i.e., 13.3 %).
- The GHG emission intensity of the chain after CO_2 is captured ranged from 3.3 % to 14.2 %, depending on the concept and transport option adopted and accounting for regional characteristics (i.e., the electricity supply mix per country). This indicates that, although small compared to the amount of CO_2 stored, the configuration of the network and in particular the emissions from the operating the batch-wise transportation are crucial in setting this intensity closer to the high range, in case of purely ship usage for direct injection, or closer to the low range if the chain is designed with intermediate storage hub and pipeline for injection.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dennis Martins Fraga reports financial support was provided by Imperial College London. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

This section contains all the parameters and assumptions considered for estimating the LCI results for batch-wise transportation and for implementing the emissions model for the MoMilp models. Table A1 summarises all the assumptions adopted for estimating the activity factor for batch-wise LCI estimates. Table A2 contains all the parameters considered in the MoMilp model implementations.

Table A1
Activity factor parameters for gate-to-gate prospective LCA impact assessment.

Element	Value		Unit
Fuel consumption ship	Capacity (kt CO_2)	Value	MWh day^{-1}
	6	248	
	10	263	
	20	301	
	50	415	
Ship maximum operation time	8,332		h $year^{-1}$
Ship speed	27.8		km $hour^{-1}$
Ship loading and unloading times	Capacity (kt CO_2)	Value	hour
	6	26	
	10	26	
	20	50	
	50	80	
Distance considered	Ships	100; 200; 400; 800	km
	Lorry	25; 50; 100	
Lorry speed	50		km $hour^{-1}$
Lorry maximum operating time	7,884		h $year^{-1}$
Lorry loading and unloading time	0.80		hour

Table A2
Emissions to air parameters for MoMILP implementation after (IEAGHG, 2004; Aspelund et al., 2006; Nie, 2009; Element Energy, 2018; Equinor, 2020).

Chain element	Type of emission	Specification				
Liquefaction	Operation	Pressure	Emission factors from electricity for liquefaction ($kgCO_2e\ t^{-1}\ CO_2$)			
			Norway	Sweden	Belgium	France	Ireland
		0.7 MPa	0.9	1.0	4.3	1.8	9.4
		1.5 MPa	0.7	0.8	3.4	1.4	7.5
Intermediate Storage	Construction	Site, storage, and workshop: 768 $tCO_2e\ kt\ ship^{-1}$				
		Administration Building: 7,067 $tCO_2e\ facility^{-1}$				
		Ship Capacity (kt)		Storage tank ($tCO_2e\ tanks^{-1}$)		
				7 MPa		15 MPa
		6		2,513		N/A
		20		5,027		
		50		12,027		
		10		2,513		5268
	Fugitive	0.0286 $tCO_2e\ h^{-1}$				
Transport - Ship	Construction	Ship Capacity (kt)		Emission ($tCO_2e\ ship^{-1}$)		
		6		4,303		
		20		14,346		
		50		35,867		
		10 for 1.5 MPa		10,061		
		10 for injection ship		7,173		
	Operation	Ship Capacity (kt)		Emission WtW ($tCO_2e\ day^{-1}$)		Energy consumption ($MWh\ trip^{-1}$)
		6		155.5		1.2
		20		189.3		2
		50		261.0		5
		10 for 1.5 MPa		165.4		2
		10 injection ship				
		Boil-off rate		0.2 % of vessel capacity after the 46th day		
Transport - Lorry	Construction	70 $tCO_2e\ lorry^{-1}$. With storage tank				
	Operation	7.7 $kgCO_2e\ km^{-1}$				
		12 $kWh\ trip^{-1}$ for loading and unloading				
Transport - Pipeline	Construction	2.2 $kgCO_2e\ kg^{-1}$steel				
	Fugitive	2.3 $tCO_2e\ km^{-1}$				
	Booster	Offshore pipeline diameter (inch)			kWh^{-1} t CO_2 transported	
		36″			2.0	
		40″			1.7	
		48″			1.4	
	Booster fugitive	23.24 t $CO_2e\ MW\ yr^{-1}$				
Injection	Construction	63,580 t $CO_2e\ well^{-1}$				
	Fugitive	2.9 g $CO_2e\ h^{-1}$				

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